Resonant Geometry Computational System, Version 2: Coupled Modes, Compact Phase Coordinates, Dynamic Coherence, and Experimental Calibration

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A Hydrogenuine research contribution, with computational assistance

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System intent. RGCS v2 unifies the geometry, timing, source-audit, and experimental work of the Vogel-cut crystal programme with three mathematical templates adapted from recent peer-reviewed physics: the compact-mode and resonance-offset architecture of Lee and Tsai [1], the anisotropy-decay functional of Gan et al. [2], and the phase-coherence measurement architecture of Koster et al. [3]. The papers are used as mathematical templates only. *Physical correspondence is determined by measurement, not assumed by terminology.*

Abstract

The Resonant Geometry Computational System version 2 (RGCS v2) is a computational and experimental framework for discrete-mode design and calibration of macroscopic faceted quartz resonators, spiral-cone geometries, and their drive systems. The framework comprises: an exact faceted-crystal geometry and density-inverse model **[EST]**; a 4096 Hz axial design ladder carrying a mandatory $\pm 5\%$ wave-speed uncertainty band and set-valued harmonic classification **[DER]**; a compact phase-coordinate mode spectrum $f_n^2 = f_b^2 + (n\kappa_\chi)^2$ with boundary-parity families, adapted as mathematical analogy from the Kaluza–Klein spectrum of Lee and Tsai [1] **[HYP]**; a signed dimensionless resonance-offset functional with Q-derived proximity classes **[DER]**; coupled-mode eigenvalue and avoided-crossing analysis **[EST]**; a dynamic amplitude–phase model with pre-registered fitting protocol **[HYP]**; a spatial phase-anisotropy functional adapting the shear-scalar form of Gan et al. [2] as a bench statistic **[DER]**; and an autocorrelation coherence metric with single-shot, phase-referenced ensemble protocols following Koster et al. [3] **[EST]**. Every numerical example, table, and figure in this manuscript is generated at build time by the open computational core `rgcs_core` 2.0.0 [4]. Fourteen pre-registered hypotheses (H-01..H-14) with explicit failure conditions, an eight-branch control-gated experimental protocol, and a four-category epistemic classification of every claim (Established / Derived / Hypothesis / Source claim) move the programme from isolated numerical coincidences to falsifiable, control-subtracted comparisons between predicted spectra and measured modes. No hypothesis in this document carries evidential status other than *untested*, and the archival negative results are preserved as such.

Contents

1 Scope and scientific classification policy

This document supersedes the v1.8 design-method manuscript [5] and the v2 prototype note. It states its epistemic boundaries once, here, and then lets the mathematics and the test programme speak.

1.1 The four categories

Every claim-bearing statement, equation, table row, and figure in this manuscript carries exactly one of four labels, typeset throughout as colored tags:

- **[EST]** **Established —** standard, independently verifiable mathematics, physics, or measurement technique. Anyone with a textbook can check it without trusting this project or its sources. Examples: the half-wave formula $f = v/2L$; 2×2 eigenvalue algebra; log-spiral arc length; the shear-scalar definition; I/Q demodulation; the autocorrelation coherence definition.
- **[DER]** **Derived —** computed *within RGCS* from explicitly stated inputs by explicitly stated (usually Established) mathematics; reproducible from the inputs; inherits the weakest label of its inputs. Examples: $L_5 = 154.052734\,\text{mm}$ given $\bar{v}_L = 6310\,\text{m/s}$ and $f_0 = 4096\,\text{Hz}$; the exact-cycle drive allocations; the engineering merit score S.
- **[HYP]** **Hypothesis —** an RGCS conjecture about the physical world that measurement could falsify. Every Hypothesis ships with a pre-registered test and failure condition (Section 15).
- **[SRC]** **Source claim —** a statement made by an external source, reported with provenance but not endorsed. This includes both the archival JH/Vogel corpus [6, 7] *and* the three reference papers' physical conclusions in their own domains (dark-matter viability, quantum-cosmological shear damping, magnon condensation) — well supported there, never physically equivalent to RGCS bench systems.

Categories are never merged; hybrid labels ("semi-established", "consistent with the source") are forbidden. A claim carrying two aspects is split: the half-wave *formula* is Established, the *choice* of $f_0 = 4096\,\text{Hz}$ is a Source claim, and the *applicability* of a 1-D model to a faceted tapered crystal is a Hypothesis (H-01a). Derived quantities inherit the weakest label in their input chain, and no amount of repetition, averaging, or plotting upgrades a label. Only a pre-registered, control-subtracted measurement campaign can move a Hypothesis toward "supported"; within this document every hypothesis is *untested* (the archival negative results are *amplitude-null, coherence-untested*; Appendix D).

1.2 Rule for the three reference papers

The mathematical identities and measurement definitions of Lee and Tsai [1], Gan et al. [2], and Koster et al. [3] are Established and are adapted here with explicit substitution maps and equation-level citations. Their model-dependent results and numbers are Source claims. No RGCS quantity is ever described with their physical vocabulary: this manuscript never asserts crystal Kaluza–Klein modes, condensation of any RGCS system, or quantum-gravitational effects on a bench. Multi-channel phase-disagreement statistics are called *spatial phase anisotropy*, never anything stronger. Each paper's non-adapted content is listed explicitly in the traceability matrix (Appendix D).

1.3 Reading aids

Equations carry their registry identifiers (RGCS-M.x), machine-checked against the computational core [4]; symbols follow the frozen notation table (Appendix A). All node coordinates use the **female-apex frame**: x runs from the female (wide) apex toward the male (narrow) apex, $0 \le x \le L$.

2 Historical and source motivation

The programme began as a practical audit of a specific archival corpus: how long a macroscopic quartz body should be for an axial half-wave at an integer multiple of 4096 Hz; how to parameterize asymmetric Vogel-style terminations; how to compute volume, mass, and density-corrected dimensions; how to make the corpus's pulse-timing prescriptions cycle-exact; and how hand or fixture loading shifts a resonance [5]. A 257-page communications log [6] was processed as a technical source corpus — layout-preserving extraction, numeric line indexing, and a curated register of findings — separating three distinct experimental systems: low-frequency node electrodes ($\sim$20 Hz, $\sim$15 V), non-contact acoustic frequency keys (1496 Hz primary), and a 4096 Hz opposed-coil generator with microsecond pulses. All of these operating points are Source claims **[SRC]**; the audit's corrections of source arithmetic (e.g., 10^{-14} A = 62,415 electrons per second, not 2,000) are Established arithmetic **[EST]**.

Version 2 exists because three recent peer-reviewed papers supply, independently, the three structures the programme lacked:

1. **A discrete-mode architecture.** Lee and Tsai [1] organize a dark-sector model around a compact coordinate: compactification generates discrete modes, boundary conditions select odd and even families, a signed dimensionless offset quantifies proximity to a two-particle threshold, and radiative corrections shift that offset. The physics is dark matter; the mathematics is a mode-spectrum-and-coupling problem that any resonator programme can implement rigorously (Sections 6–8).
2. **A decay functional for directional disagreement.** Gan et al. [2] show that in a modified loop-quantum-cosmology model the anisotropic shear scalar decays exponentially after the bounce. RGCS adapts the *shear-scalar form* and the *model-comparison methodology* as a testable bench statistic for multi-channel phase data (Section 11); nothing cosmological is imported — no crystal produces a cosmological bounce, and their Planck-unit decay coefficient is never reused.
3. **A coherence measurement architecture.** Koster et al. [3] demonstrate phase-referenced single-shot detection, random ensemble initial phase, threshold behavior, and an amplitude-independent autocorrelation coherence metric in a parametrically pumped magnon system. RGCS adopts the measurement chain, the ensemble statistics, and the metric (Sections 10, 13); the magnon physics stays theirs.

The useful move is not to rename a crystal after any of these systems. It is to implement the same disciplined questions: What coordinate is closed? What discrete modes do the boundary conditions allow? Which are symmetric or antisymmetric? How close are two mode families to a specified relationship, in units of measured linewidth? Does coupling produce an avoided crossing? Which corrections are required before any of this predicts a real measurement — and what observation would prove the whole construction wrong?

3 Established geometric and wave foundations

3.1 State vector

The full system state is the typed tuple RGCS-M.0 **[DER]**

$$\Psi(t) = \big(\mathcal{G},\ \mathcal{S}_{sp},\ \mathcal{B},\ \{\omega_n\},\ \{Q_n\},\ \{u_n\},\ \{z_n(t)\},\ \chi(t),\ \Lambda,\ \Sigma_\phi(t),\ \mathcal{C}_w(t),\ \mathcal{O}(t)\big), \tag{1}$$

where $\mathcal{G}$ is the frozen crystal geometry and material state (including the wave speed as an uncertain parameter), $\mathcal{S}_{sp}$ the spiral state, $\mathcal{B}$ the boundary/parity state, $\{\omega_n, Q_n, u_n\}$ the modal backbone, $\{z_n(t)\}$ the complex modal amplitudes, $\chi(t)$ the compact phase coordinate, Λ the measured drive–mode overlap, $\Sigma_\phi(t)$ the phase-anisotropy tensor, $\mathcal{C}_w(t)$ the coherence trace, and $\mathcal{O}(t)$ the measured observables with their controls, artifact register, and negative-result ledger. Measurement is part of the state; controls are not an afterthought. Figure 1 shows the module architecture and the calibration loop.

3.2 Faceted crystal geometry

For an N_f-sided regular section of diameter D measured across vertices RGCS-M.1 **[EST]**

$$A(D) = \frac{N_f}{8}\, D^2 \sin\frac{2\pi}{N_f}, \qquad r_a = \frac{D}{2}\cos\frac{\pi}{N_f}, \tag{2}$$

with the across-flats convention $A(D) = N_f(D/2)^2\tan(\pi/N_f)$, $r_a = D/2$ RGCS-M.2–M.3. Termination (cap) heights follow the declared angle convention RGCS-M.4 **[EST]**:

$$h = r_a\tan\alpha\ (\texttt{face\_slope}), \qquad h = \frac{r_a}{\tan\alpha}\ (\texttt{axis\_to\_face}), \qquad h = \frac{r_a}{\tan(\alpha/2)}\ (\texttt{apex\_included}).$$

The convention must be declared with every dimension set; the default angle values $51.843^\circ/60^\circ$ are Source claims **[SRC]** (their numerological audit is Established arithmetic: $\arctan\sqrt{\varphi} = 51.827292^\circ$, a nonzero

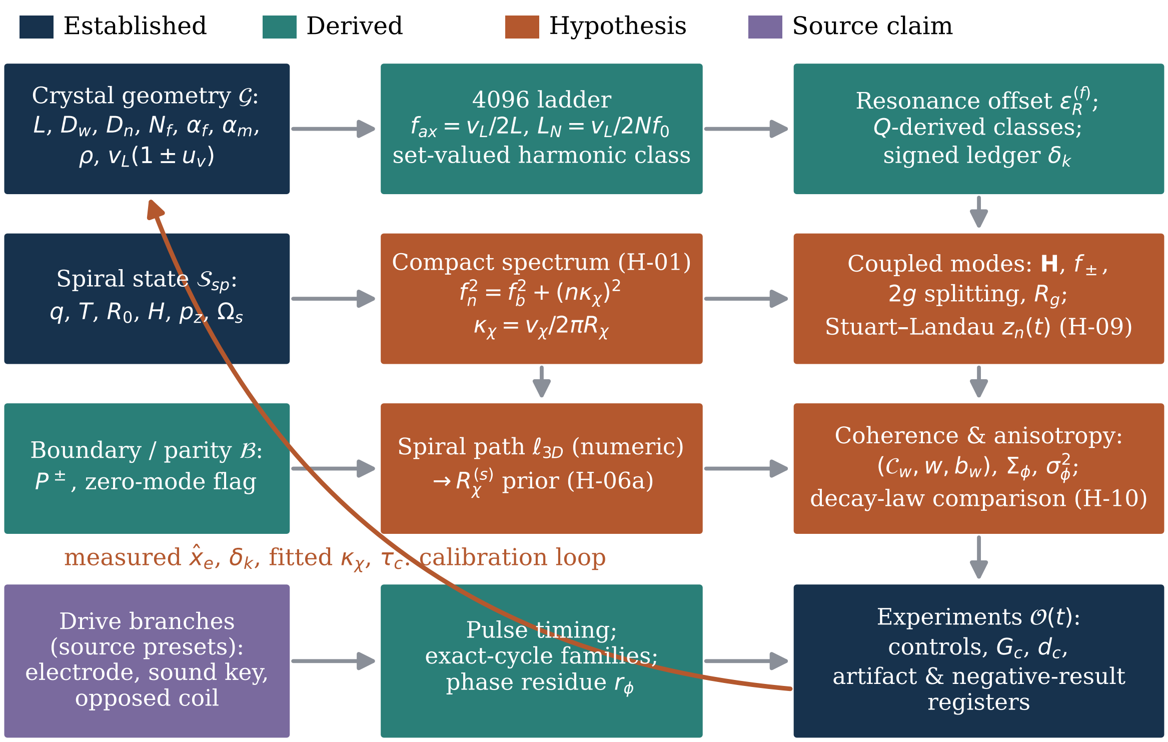

Figure 1: RGCS v2 system architecture. Boxes are `rgcs_core` module groups, colored by the classification of their central content; the return path carries measured quantities (node coordinate $\hat{x}_e$, correction ledger δ_k, fitted κ_χ and τ_c) back into the forward model.

$-0.015708°$ from the proposed value — the manuscript makes no golden-ratio equality claim). Total volume is the tapered polygonal frustum plus two pyramidal caps RGCS-M.5 **[EST]**,

$$V = \frac{h_s}{3}\left(A_w + A_n + \sqrt{A_w A_n}\right) + \frac{A_w h_f}{3} + \frac{A_n h_m}{3}, \qquad h_s = L - h_f - h_m > 0, \tag{3}$$

and $m = \rho V$ RGCS-M.6 with handbook density $\rho = 2.65\,\mathrm{g/cm^3}$ for α-quartz [8]. The density inverse — solving $\rho V(s_D D_w, s_D D_n; L) = m^*$ for the diameter scale s_D given a measured mass — has a unique root because V is strictly increasing in s_D; the core solves it by guarded Newton iteration to $|\Delta m|/m^* < 10^{-10}$ RGCS-M.7 **[DER]**.

3.3 Wave propagation and the uncertain wave speed

α-quartz is trigonal and anisotropic [8–10]: from Bechmann's elastic constants, the pure longitudinal speed along the crystallographic Z axis is $\sqrt{c_{33}/\rho} \approx 6320\,\mathrm{m/s}$, while along X it is $\sqrt{c_{11}/\rho} \approx 5720\,\mathrm{m/s}$ **[EST]**. The corpus value $\bar{v}_L = 6310\,\mathrm{m/s}$ is consistent with the Z-axis longitudinal speed [9] but was imported uncited **[SRC]**; RGCS therefore treats the wave speed as a first-class uncertain parameter RGCS-M.10 **[DER]**:

$$v_L = \bar{v}_L(1 + \delta_v), \qquad \mathbb{E}[\delta_v] = 0, \qquad \mathrm{sd}(\delta_v) = u_v = 0.05, \tag{4}$$

where the default u_v covers the X–Z longitudinal spread until a specimen's crystallographic orientation is measured, at which point u_v may be reduced per specimen with cited constants. Since every ladder frequency and length below is proportional to v_L, the relative uncertainty propagates unchanged RGCS-M.11: $u(f_{ax})/f_{ax} = u(L_N)/L_N = u_v$. **Interpretation rule:** any sub-percent "match" to a ladder frequency is two orders of magnitude smaller than this systematic band and is Derived-from-Hypothesis arithmetic, never confirmation (Section 4).

Table 1: Set-valued harmonic classification (RGCS-M.12) at $u_v = 0.05$, $f_0 = 4096$ Hz. Fractional error is quoted against $N_{\text{nearest}} f_0$; by the interpretation rule (RGCS-M.11, D-04) every value here is Derived-from-Hypothesis arithmetic, never confirmation.

L (mm)	f_{ax} (Hz)	$\pm 1\sigma$ band (Hz)	N_{eff}	nearest N	class set $\mathcal{N}$	ambiguous	error
154.052734	20480.0	[19456, 21504]	5.0000	5	{5}	no	+0.000%
110.000000	28681.8	[27248, 30116]	7.0024	7	{7}	no	+0.034%
116.000000	27198.3	[25838, 28558]	6.6402	7	{6, 7}	yes	-5.140%

4 Crystal geometry and the 4096 design ladder

The first-order axial half-wave estimate is RGCS-M.8

$$f_{ax} = \frac{v_L}{2L}, \qquad L_N = \frac{v_L}{2N f_0}, \tag{5}$$

where the formula is Established **[EST]**, the choice $f_0 = 4096\,\text{Hz}$ is a Source claim **[SRC]** [6], and the applicability of a pure longitudinal 1-D model to a faceted, tapered, hand-finished crystal is Hypothesis H-01a **[HYP]** (cf. the extensional-mode literature for uniform cylinders [11]). For $\bar{v}_L = 6310\,\text{m/s}$ the ladder constant is $770.263671875/N$ mm, giving $L_5 = 154.052734$ mm and $L_7 = 110.037667$ mm, each carrying $\pm 5\%$.

The 110 mm specimen, stated correctly. A source-referenced 110 mm specimen yields $f_{ax} = 28681.818 \pm 1434\,\text{Hz}$ (1σ), which sits $+9.818$ Hz ($+0.0342\%$) above $7 \times 4096 = 28672$ Hz. The celebrated 0.03% agreement is real arithmetic — and it lives *inside* a $\pm 5\%$ systematic band roughly $170\times$ wider than the residual. It is classified Derived-from-Hypothesis-and-Source-claim **[DER]**: it rests on the 1-D applicability hypothesis (H-01a) and on the source-claimed specimen length, and it is never presented as confirmation of anything. What it does earn is a pre-registered measurement: the specimen's actual free fundamental, against the full interval prediction.

Set-valued harmonic classification. Because the prediction is an interval, classification must be too RGCS-M.12 **[DER]**:

$$\mathcal{N} = \Big\{ N' \in \mathbb{N} : \big[f_{ax}(1-u_v),\ f_{ax}(1+u_v)\big] \cap \big[(N' - \tfrac{1}{2})f_0,\ (N' + \tfrac{1}{2})f_0\big] \neq \emptyset \Big\}. \tag{6}$$

A 116 mm example gives $f_{ax} = 27198.3$ Hz, $N_{\text{eff}} = 6.6402$, and the *ambiguous* class set $\mathcal{N} = \{6, 7\}$ at $u_v = 0.05$: the specimen cannot honestly be assigned a single harmonic number from length alone (Table 1).

5 Spiral and conical geometry and the compact phase coordinate

The spiral-cone state path is the four-component curve RGCS-M.30 **[EST]** (differential geometry; physical relevance is Hypothesis H-06):

$$X(\theta) = \begin{bmatrix} R_0 e^{-a\theta} \cos\theta \\ R_0 e^{-a\theta} \sin\theta \\ H\big(1 - (r/R_0)^{p_z}\big) \\ \chi_0 + \Omega_s\,\theta \end{bmatrix}, \qquad \theta \in [0,\ 2\pi T], \qquad a = \frac{\ln q}{2\pi}, \tag{7}$$

with growth ratio q per turn, T turns, outer radius R_0, cone height H, and cone exponent p_z (the z-law is equivalent to $H(1 - e^{-p_z a\theta})$). The fourth coordinate reduced modulo 2π is the compact phase χ — a modeling construct, *not* a spatial extra dimension. For $q = e$: pitch parameter $a = 0.15915494$ RGCS-M.31, planar scale-rotation eigenvalue $\lambda_s = -a + i$ RGCS-M.32, and curvature invariant $r\kappa = 1/\sqrt{1+a^2} = 0.98757049$ RGCS-M.33, all **[EST]**.

Path length: the numeric value is authoritative. The exact planar arc length is $\ell_{pl} = R_0\sqrt{1+a^2}\,(1 - e^{-2\pi a T})/a = 374.744\,\text{mm}$ RGCS-M.34 **[EST]**. The *project-defining* 3-D path length is the converged numeric polyline RGCS-M.35 **[DER]**,

$$\ell_{3D} = \lim_{\text{refinement}} \sum_i \big\| X_{3D}(\theta_{i+1}) - X_{3D}(\theta_i) \big\| = 384.022\ \text{mm}, \tag{8}$$

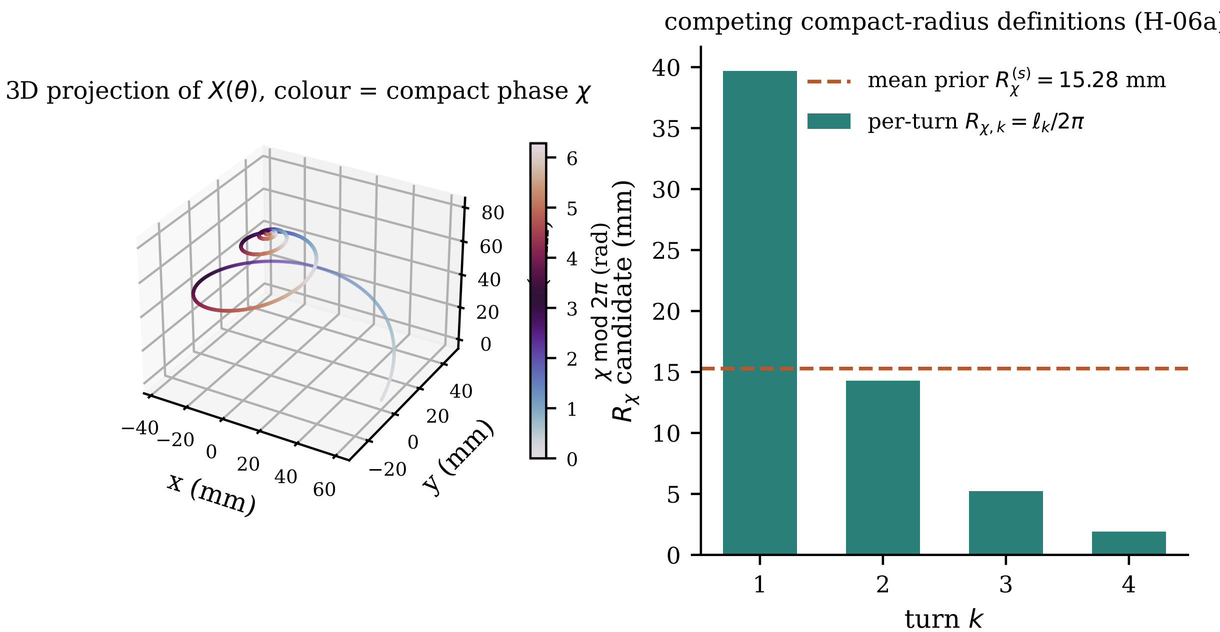

Figure 2: Left: three-dimensional projection of the four-component path $X(\theta)$ RGCS-M.30 for $q = e$, $T = 4$, $R_0 = 60$ mm, $H = 80$ mm, $p_z = 1.5$, colored by the compact phase coordinate χ mod 2π. Right: the two competing compact-radius definitions; the per-turn radii contract by $\approx 1/q$ per turn, so the mean prior hides real structure that the model comparison of H-06a must adjudicate.

computed with sample doubling until the relative change is below 10^{-6}. The workbook closed form $\ell_{pl}\sqrt{1 + (H/\ell_{pl})^2}$ is *retired as a definition*: it assumes the climb is distributed uniformly along the planar length and here evaluates to 383.188 mm, an error of -0.22% against ℓ_{3D} — it may appear only as this labeled approximation with its error printed.

Compact-radius priors. Two competing definitions bridge the spiral to the compact spectrum, both Hypothesis-classified **[HYP]**:

$$R_\chi^{(s)} = \frac{\ell_{3D}}{2\pi T} = 15.280 \text{ mm} \quad \text{RGCS-M.36}, \qquad R_{\chi,k} = \frac{\ell_k}{2\pi} = 39.68, 14.30, 5.22, 1.92 \text{ mm} \quad \text{RGCS-M.37}, \tag{9}$$

the mean-per-turn prior (averaging assumption A-07) and the per-turn alternative whose lengths contract by $\approx 1/q$ each turn (Figure 2). The pre-registered model comparison of Section 14 must test both against measured mode spacing; neither is presumed. The prototype default "$R_\chi = 100$ mm" is an arbitrary placeholder unrelated to these spiral values, and the code never presents them as consistent.

6 Compact-coordinate mode spectrum

6.1 The spectrum and its substitution map

RGCS v2 posits a closed wave path of effective radius R_χ supporting propagation at speed v_χ, with mode spectrum RGCS-M.13 **[HYP]** (Hypothesis H-01, built on Established Kaluza–Klein mode algebra):

$$f_n^2 = f_b^2 + \left(\frac{n\, v_\chi}{2\pi R_\chi}\right)^2. \tag{10}$$

Equation (10) is a direct mathematical adaptation of the tree-level dark-photon mass formula of Lee and Tsai [1, Eqs. (9)–(10)], $m_{\gamma_D}^{(n)} = \sqrt{m_B^2 + n^2/R^2}$, under the explicit substitution map

$$m \longrightarrow f, \qquad m_B \longrightarrow f_b, \qquad \frac{1}{R} \longrightarrow \frac{v_\chi}{2\pi R_\chi},$$

Table 2: Compact-coordinate spectrum for $f_b = 0$, $\kappa_\chi = 10042.68$ Hz $\pm 5\%$ ($v_\chi = \bar{v}_L$, $R_\chi = 100$ mm placeholder, D-21). The $n = 0$ member is excluded unconditionally at $f_b = 0$ (RGCS-M.17). Intervals are the mandatory $\pm u_v$ propagation of RGCS-M.15.

n	parity	f_n (Hz)	$\pm 1\sigma$ band (Hz)
1	odd	10042.68	[9541, 10545]
2	even	20085.35	[19081, 21090]
3	odd	30128.03	[28622, 31634]
4	even	40170.71	[38162, 42179]
5	odd	50213.38	[47703, 52724]
6	even	60256.06	[57243, 63269]

which is dimensionally exact for wave propagation around a circumference $2\pi R_\chi$. The massless-base limit $f_b = 0$ reproduces their fermion tower $m_E^{(n)} = n/R$ [1, Eq. (9)]. The physical claim — that any measured RGCS mode family obeys Eq. (10) better than simpler modal models on held-out modes — is Hypothesis H-01 with a pre-registered failure condition (Section 15). RGCS's χ is a closed wave path; nothing here asserts an extra spatial dimension of a crystal.

6.2 Identifiability: report κ_χ, not (v_χ, R_χ)

Only the combination RGCS-M.14 **[DER]**

$$\kappa_\chi \equiv \frac{v_\chi}{2\pi R_\chi} \qquad [\text{Hz}] \tag{11}$$

enters Eq. (10): v_χ and R_χ are *not separately identifiable* from mode spacing alone. Fits report κ_χ with uncertainty; R_χ is quoted only when v_χ is independently fixed (e.g., $v_\chi := v_L$ with its u_v), and then $u(R_\chi)/R_\chi \geq u_v$ necessarily. The spiral prior of Section 5 breaks the degeneracy only under Hypothesis H-06a. With $v_\chi = \bar{v}_L$ and the placeholder $R_\chi = 100$ mm, $\kappa_\chi = 10042.68$ Hz and the $n = 1$ mode is the golden value 10042.6769 Hz $\pm 5\%$; with the spiral prior $R_\chi^{(s)} = 15.280$ mm instead, $\kappa_\chi \approx 65725$ Hz — the two differ by a factor of $\sim$6.5, which is precisely why the placeholder is labeled a placeholder. First-order uncertainty propagation RGCS-M.15 makes every spectrum output an interval (Table 2, Figure 3).

6.3 Boundary parity and the zero-mode rule

On a compact coordinate with two distinguished boundary points, mode functions divide into parity families RGCS-M.16 **[EST]** (as index-set definitions; their physical relevance to RGCS resonators is Hypothesis H-03):

$$u_n^-(\chi) \propto \sin\frac{n\chi}{2},\ n \in \mathbb{I}^- = \{1, 3, 5, \dots\}, \qquad u_n^+(\chi) \propto \cos\frac{n\chi}{2},\ n \in \mathbb{I}^+ = \{0, 2, 4, \dots\}, \tag{12}$$

adapted from the boundary-condition classification of Lee and Tsai [1, Table I, Eqs. (1)–(2)], where orbifold conditions select odd fermion and even dark-photon modes. In RGCS, odd means an antisymmetric displacement/current/pressure pattern about the reference plane and even means symmetric; assignment is made by two-sensor phase or finite-element mode shape, *never* by which family contains a desired frequency (Figure 4).

The zero mode requires care RGCS-M.17 **[DER]**. The odd family never contains $n = 0$ ($\sin \equiv 0$ is not a mode). The even family contains $n = 0$ mathematically, with $f_{n=0} = f_b$; in the strict Lee–Tsai template a zero-flux condition removes the mediator zero mode [1, Eq. (2)]. RGCS therefore exposes an explicit boolean `include_zero_mode` on the even family: `True` (default when $f_b > 0$) lists the physical base mode; `False` reproduces the strict template for exact-analogy tests. When $f_b = 0$ the $n = 0$ member is a DC pattern and is excluded unconditionally. **Every spectrum in this manuscript that cites Lee and Tsai [1] states its flag** (all figures and tables here use `include_zero_mode = True` except where marked), and a measured low-lying symmetric mode where the strict template forbids one is pre-registered evidence that the analogy is partial (H-04).

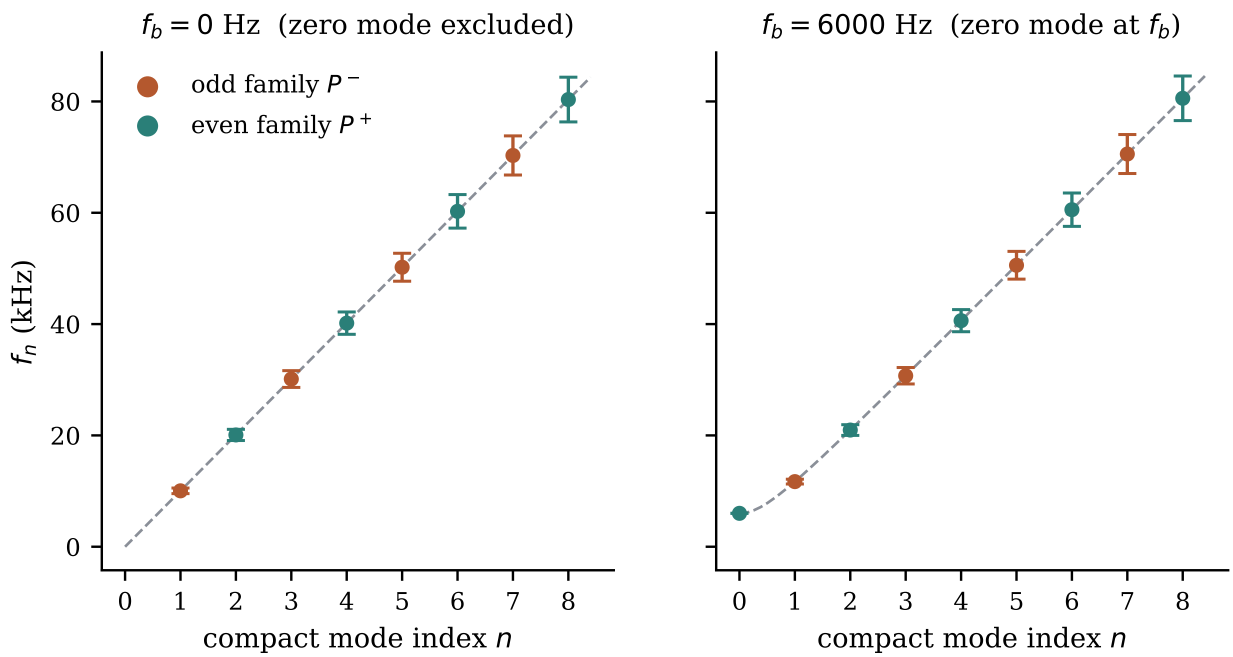

Figure 3: Compact-mode ladder RGCS-M.13 with mandatory $\pm u_v$ uncertainty bands, for a massless base ($f_b = 0$, left; linear in n, the Lee and Tsai [1] fermion-tower limit, zero mode excluded unconditionally) and a finite base frequency ($f_b = 6$ kHz, right; quadrature form with the $n = 0$ member at f_b). Colors distinguish the odd and even parity families of Section 6.3.

7 Resonance-offset functional

7.1 Definition and sign convention

For a measured bridge (mediator-like) mode f_m and target (matter-like) mode f_x with pair multiple p (default 2), RGCS defines RGCS-M.18 **[DER]**

$$\epsilon_R^{(f)} = \frac{f_m^2 - (pf_x)^2}{(pf_x)^2}, \tag{13}$$

with sign convention identical to the resonance level of Lee and Tsai [1, Eq. (13)]: $\epsilon_R^{(f)} > 0$ places the bridge above the threshold, $\epsilon_R^{(f)} < 0$ below. Their invisible/visible dark-photon interpretation of the sign has *no* RGCS analogue. The conjecture that transfer or splitting enhancement localizes near $\epsilon_R^{(f)} = 0$ is Hypothesis H-02 **[HYP]**. For small detunings, $\epsilon_R^{(f)} = (1 + d_{\text{lin}})^2 - 1 \approx 2d_{\text{lin}}$ with $d_{\text{lin}} = f_m/(pf_x) - 1$ RGCS-M.19 **[EST]**.

7.2 Q-derived proximity classes

Earlier drafts borrowed the window $10^{-8} \le \epsilon_R \le 10^{-4}$ from the dark-sector context [1]; that window is astrophysics, not metrology, and is retired. RGCS classes now derive from measured linewidths RGCS-M.20 **[DER]**: a mode pair with effective quality factor $Q_{\text{eff}} = 2/(1/Q_m + 1/Q_x)$ has fractional half-linewidth scale

$$\epsilon_Q \equiv \frac{1}{Q_{\text{eff}}}, \tag{14}$$

and the classes are WITHIN LINEWIDTH ($|\epsilon_R^{(f)}| \le \epsilon_Q$), NEAR ($\le 5\epsilon_Q$), MODERATE ($\le 50\epsilon_Q$), and FAR otherwise. For $Q_m = 1000$, $Q_x = 800$: $\epsilon_Q = 1.1250 \times 10^{-3}$ — resolvable by ordinary instrumentation, unlike the retired bins. A resonance-class string reported without $u(\epsilon_R^{(f)})$ is non-compliant (Table 3, Figure 5). The far-detuned reference $\epsilon = 1.25$ — the value at which the Lee–Tsai mediator sits at three times the pair threshold [1] — is retained *only* as the convention defining a deliberately non-resonant control configuration; it is a control convention, not physics.

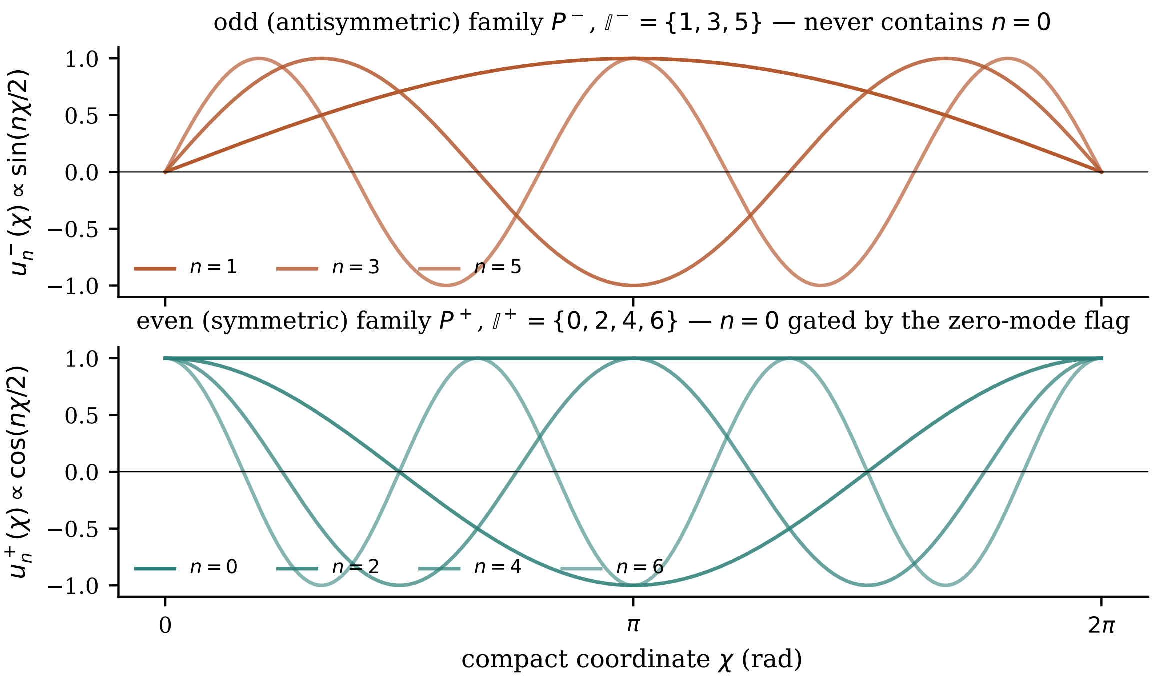

Figure 4: Parity families RGCS-M.16 on the compact coordinate, adapted from Lee and Tsai [1, Table I]. Zero-mode flag: `include_zero_mode = True` (the $n = 0$ even member is drawn; under the strict Lee–Tsai template it would be removed by the zero-flux condition).

Table 3: Q-derived resonance classes (RGCS-M.20) for a target mode $f_x = 20480$ Hz, $p = 2$, $Q_m = 1000$, $Q_x = 800$ ($\epsilon_Q = 1/Q_{\text{eff}} = 1.125 \times 10^{-3}$). A class string without $u(\epsilon_R)$ is non-compliant (policy §3.4). Sweep spans include the six-linewidth floor of RGCS-M.21 (engineering heuristic).

f_m (Hz)	$\epsilon_R^{(f)}$	$u(\epsilon_R)$	class	sweep span (Hz)
40960	+0.00000	2×10^{-4}	within linewidth	246
40970	+0.00049	2×10^{-4}	within linewidth	246
41100	+0.00685	2×10^{-4}	moderate	560
41800	+0.04144	3×10^{-4}	moderate	3360
45000	+0.20699	5×10^{-4}	far	16160
61440	+1.25000	1×10^{-3}	far	81920

7.3 Corrections shift the offset — with sign

The Lee–Tsai one-loop corrections [1, Eqs. (11)–(12)] carry definite signs and can move ϵ_R through zero; RGCS's calibration analogue is the signed correction ledger RGCS-M.22, M.45 **[DER]**

$$f_{i,\text{corr}} = f_{i,0}\Big(1 + \sum_k \delta_k^{(i)}\Big), \qquad u(\epsilon_R^{(f)})\big|_{\epsilon\approx 0} \approx 2\sqrt{\left(\frac{u(f_m)}{f_m}\right)^2 + \left(\frac{u(f_x)}{f_x}\right)^2}, \tag{15}$$

with $\delta_k \in \{\delta_{\text{geometry}}, \delta_{\text{anisotropy}}, \delta_{\text{loading}}, \delta_T, \delta_{\text{fixture}}, \delta_{\text{drive}}\}$ each signed and carrying its own uncertainty. A worked ledger example (loading -0.4%, temperature $+0.08\%$ on f_m; fixture $+0.1\%$ on f_x) moves an exactly-zero offset to $\epsilon_R^{(f)} = -0.00837 \pm 0.00017$: through zero and out the other side. Repeated δ_k measurements must be stable across calibrations before any $\epsilon_R^{(f)}$ is interpreted — this gates Hypothesis H-02.

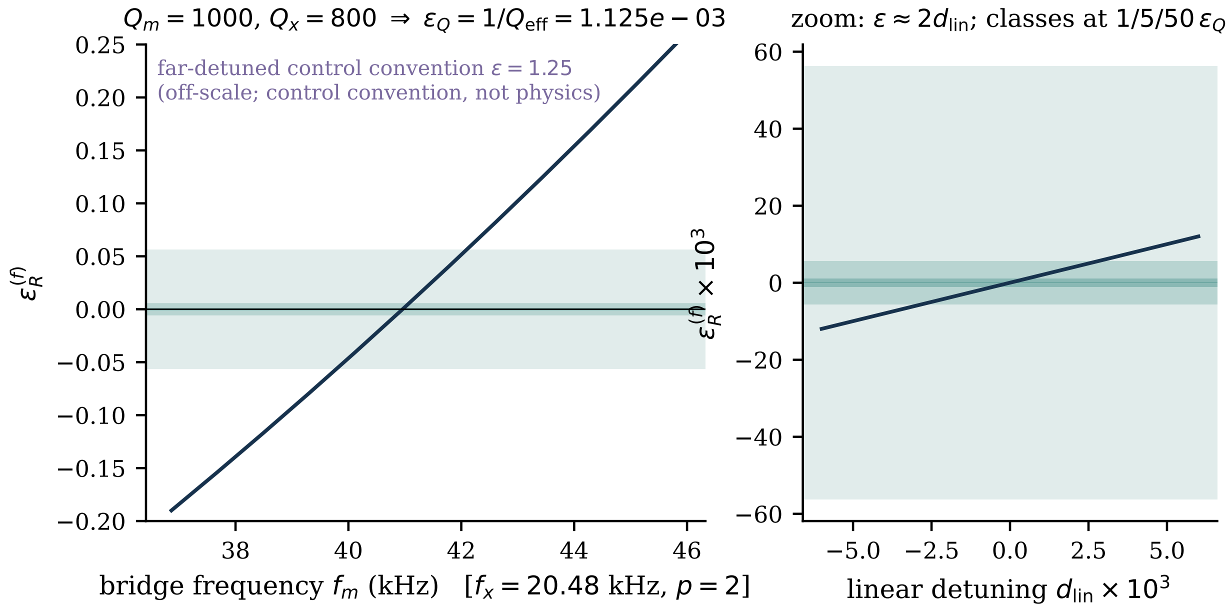

Figure 5: Resonance-offset functional RGCS-M.18 and Q-derived classes RGCS-M.20 for $f_x = 20480\,\mathrm{Hz}$, $p = 2$, $Q_m = 1000$, $Q_x = 800$. Shaded bands mark the WITHIN-LINEWIDTH, NEAR, and MODERATE classes at $1/5/50 \times \epsilon_Q$; sign convention follows Lee and Tsai [1, Eq. (13)].

8 Coupled eigenmodes and avoided crossings

Two modes with uncoupled frequencies f_A, f_B and coupling g obey the standard symmetric two-mode model RGCS-M.23–M.25 **[EST]** [12]:

$$\mathbf{H} = \begin{bmatrix} f_A & g \\ g & f_B \end{bmatrix}, \qquad f_\pm = \frac{f_A + f_B}{2} \pm \sqrt{\left(\frac{f_A - f_B}{2}\right)^2 + g^2}, \qquad \vartheta_{\mathrm{mix}} = \tfrac{1}{2}\,\mathrm{atan2}(2g,\ f_A - f_B). \tag{16}$$

At zero detuning the splitting is $2g$ (golden value: $f_A = f_B = 1000\,\mathrm{Hz}$, $g = 10\,\mathrm{Hz} \Rightarrow 990/1010\,\mathrm{Hz}$). With FWHM linewidths $\Gamma_n = f_n/Q_n$ RGCS-M.26 the strong-coupling criterion is RGCS-M.27 **[EST]** [12, 13]

$$R_g = \frac{2g}{\Gamma_A + \Gamma_B} \gtrsim 1 \quad \Longleftrightarrow \quad \text{splitting resolved against the combined linewidth.} \tag{17}$$

Figure 6 shows the worked case $f_B = 20480\,\mathrm{Hz}$, $g = 25\,\mathrm{Hz}$, $Q = 1000$: splitting $2g = 50\,\mathrm{Hz}$ against $\Gamma = 20.48\,\mathrm{Hz}$ per mode, $R_g = 1.22$ — just resolved. A resolved anticrossing is a far stronger signature than a single coincident peak. The N-mode generalization RGCS-M.28 is a real symmetric eigenproblem (orthogonal eigenvectors, `numpy.linalg.eigh`), valid in the near-degenerate weak-coupling regime; it is *not* a substitute for full elastodynamics.

Geometry scaling of the coupling. The Lee–Tsai dimensional reduction $g_1 = g_1^{(5)}/\sqrt{L}$ [1, Eq. (6)] motivates the testable scaling RGCS-M.29 **[HYP]** (H-05)

$$g \propto (2\pi R_\chi)^{-1/2} \quad \text{at fixed transducer and mode pair;} \tag{18}$$

a fitted g that fails to scale as $\sqrt{R_2/R_1}$ after a controlled compact-path change kills this analogy row.

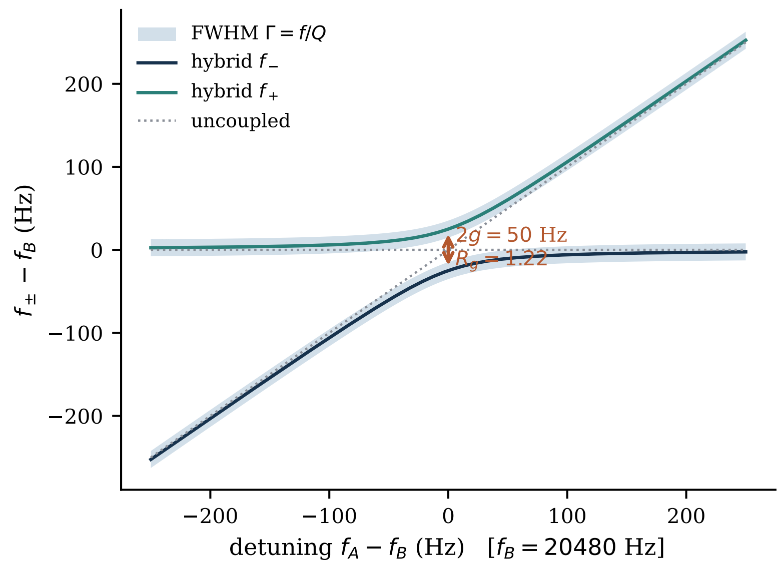

Figure 6: Avoided crossing RGCS-M.24 for a fixed mode at 20480 Hz coupled with $g = 25$ Hz to a swept mode ($Q_A = Q_B = 1000$). Shaded bands are the FWHM linewidths; the minimum splitting $2g$ against the combined linewidth gives $R_g = 1.22$.

9 Node, loading, damping, and correction models

9.1 Node coordinate (female-apex frame)

The corpus distinguishes a functional ("vorticial") node from the metric center **[SRC]**. RGCS stores three quantities in the declared frame (x from the female apex) RGCS-M.38–M.40:

$$x_m = \frac{L}{2}, \qquad x_g = h_f + \frac{h_s}{2} = \frac{L + h_f - h_m}{2}, \qquad \hat{x}_e = \text{measured node coordinate}, \tag{19}$$

where x_g **[DER]** applies the free–free uniform-bar midpoint principle to the prismatic shaft and is a *prior only*, and a measured $\hat{x}_e$ always takes precedence. For the default $N = 5$ geometry ($L = 154.052734$ mm, $h_f = 17.415434$ mm, $h_m = 14.812763$ mm): $x_g = 78.3277$ mm versus $x_m = 77.0264$ mm. The historical 75.725 mm estimate is exactly $L - x_g = 75.7250$ mm — the same point expressed from the male apex, a frame conversion, not a competing estimator; a third legacy estimator with no derivation was deleted from the code. The node-alignment factor $N_x = \exp[-((x_d - x_*)/\sigma_x)^2]$ RGCS-M.41 is an engineering heuristic feeding only the merit score, never evidence.

9.2 Loading

The loading factor is the measured ratio RGCS-M.42 **[EST]** $k_H = f_{\text{loaded}}/f_{\text{free}}$, a long-standing quartz measurement problem [10, 14]. Under Hypothesis H-08 — that loading acts as pure added modal mass with modal fraction η — first-order SDOF algebra gives RGCS-M.43

$$\Delta M_H = M_{\text{eff}}\left(\frac{1}{k_H^2} - 1\right), \qquad M_{\text{eff}} = \eta\, m. \tag{20}$$

Golden example: $k_H = 0.9866751$ (the $152/154.052734$ ratio), $m = 154$ g, $\eta = 0.5 \Rightarrow \Delta M_H = 2.0938$ g — conditioned on the *unmeasured* η, whose scale degeneracy with ΔM_H requires either fixing η or a two-loading protocol. The length-shortfall proxy $\tilde{k}_H = L_{\text{actual}}/L_{\text{target}}$ RGCS-M.44 **[HYP]** (H-08b) is a *distinct* quantity: it equals k_H only if $f \propto 1/L$ exactly and loading is intended to close the gap; code and text keep the tilde.

Table 4: Spiral-path and node-coordinate worked example. Spiral defaults $q = e$, $T = 4$, $R_0 = 60$ mm, $H = 80$ mm, $p_z = 1.5$; crystal $L = 154.052734$ mm, $D_w/D_n = 32/25$ mm, $N_f = 6$, $\alpha_f/\alpha_m = 51.843°/60°$ (angle values are Source claims); node rows use the corpus default cap heights $h_f = 17.415434$ mm, $h_m = 14.812763$ mm. Node coordinates are in the female-apex frame (x measured from the wide apex).

quantity	value (live)	status
pitch parameter $a = \ln q/2\pi$	0.159154943	Established
curvature invariant $r\kappa$	0.987570492	Established
exact planar arc length ℓ_{pl}	374.744 mm	Established
numeric 3D path length ℓ_{3D} (authoritative)	**384.022 mm** (converged to 10^{-6})	Derived
retired closed form $\ell_{pl}\sqrt{1+(H/\ell_{pl})^2}$	383.188 mm (-0.22% vs ℓ_{3D})	labelled approximation
mean compact-radius prior $R_\chi^{(s)} = \ell_{3D}/2\pi T$	15.280 mm	Hypothesis (A-07)
per-turn radii $R_{\chi,k} = \ell_k/2\pi$	39.68, 14.30, 5.22, 1.92 mm	Hypothesis (alt.)
metric centre $x_m = L/2$	77.0264 mm	Established
node prior $x_g = (L + h_f - h_m)/2$ (female frame)	78.3277 mm	Derived (prior only)
same point, male frame $L - x_g$	75.7250 mm	frame conversion

9.3 Damping and the correction ledger

Modal damping follows from measured quality factors RGCS-M.49 **[EST]**: $\gamma_n = \omega_n/(2Q_n)$ (amplitude decay; energy decays at $2\gamma_n$). The full measurement model RGCS-M.45, M.59 is

$$f_{\text{meas}} = f_{0,\text{pred}}\Big(1 + \textstyle\sum_k \delta_k\Big) + \varepsilon_{ns}, \tag{21}$$

first-order additive (a nonlinear or Bayesian treatment supersedes it when $|\sum \delta| > 0.02$), with independent Gaussian noise ε_{ns}; fits minimize the standard negative log-likelihood with bootstrap checks at small n [15].

10 Dynamic coherence and phase evolution

Version 1 was static. Version 2 adds the time axis: how a drive populates modes, how phase order can (or cannot) emerge, and how amplitude and phase order decay — with one normative rule imported from the measurement architecture of Koster et al. [3]:

> **Amplitude and coherence are separate observables.** High amplitude is not evidence of phase order, and phase order can outlive detectable amplitude. Every dynamic RGCS measurement reports both, and every coherence value ships as the triplet $(\mathcal{C}_w, w, b_w)$: the value, its analysis window, and the measured noise baseline for that window. **[EST]** (metric property).

10.1 Amplitude–phase dynamics

The modal state evolves under the canonical complex system RGCS-M.46 **[HYP]** (H-09; the mathematical structure — the slowly-varying-envelope / Stuart–Landau normal form near a Hopf bifurcation — is Established [16, 17]):

$$\dot{z}_n = (G_n - \gamma_n + i\omega_n)\, z_n - \sum_m \beta_{nm}|z_m|^2 z_n + \sum_m K_{nm}\, z_m, \tag{22}$$

with drive gain G_n, damping $\gamma_n = \omega_n/2Q_n$, cubic saturation β_{nm}, and linear inter-mode coupling K_{nm} — the time-domain counterpart of the frequency-domain g_{nm} is the anti-Hermitian coupling $K_{nm} = i\,2\pi g_{nm}$ (g in Hz): at degeneracy the pair eigenvalues are $i(\omega_0 \pm 2\pi g)$, producing spectral peaks at $f_0 \pm g$ — the $2g$ splitting of Sec. 8 — whereas a real-symmetric K of equal magnitude splits growth rates, not frequencies. The pre-registered consistency requirement is $|K_{nm}| = 2\pi g_{nm}$: disagreement between time-domain and frequency-domain coupling estimates beyond 2σ is an internal-inconsistency flag. The polar decomposition RGCS-M.47–M.48 gives the amplitude and phase equations

$$\dot{A}_n = (G_n - \gamma_n)A_n - \sum_m \beta_{nm}A_m^2 A_n + \sum_m K_{nm}A_m\cos(\phi_m - \phi_n), \qquad \dot{\phi}_n = \omega_n + \sum_m K_{nm}\frac{A_m}{A_n}\sin(\phi_m - \phi_n), \tag{23}$$

integrated in the complex form (the polar form is singular as $A \to 0$). Registered alternatives (full linear equations, Van der Pol gain, sin-bounded saturation, parametric $f/2$ terms for the parametric drive branch only, Kuramoto phase-only) are documented with the reasons each is not the default; the linear null $\beta = K = 0$ is the model-comparison baseline for H-09.

10.2 The four-stage template and spontaneity

Koster et al. [3] observe four temporal stages in a parametrically pumped magnon gas: incoherent pair injection with no global phase (high amplitude, baseline coherence); nonlinear redistribution; emergence of per-run-stable phase *with ensemble-uniform phase across 1,000 runs* — order that is spontaneous, not drive-imprinted; and free post-drive evolution in which coherence persists below the amplitude noise floor. RGCS adopts the *stage structure* as an analysis template (their physics stays theirs **[SRC]**): stage labels are assigned only from the measured (amplitude, coherence, ensemble-phase) triple, never assumed. The spontaneity test is the Rayleigh statistic RGCS-M.61 **[EST]** [18] on one phase per run,

$$\bar{R} = \Big|\frac{1}{n_s}\sum_{\text{shots}} e^{i\phi_{\text{shot}}}\Big|, \qquad Z_R = n_s\bar{R}^2 : \tag{24}$$

uniform ensemble phase with high per-shot coherence indicates spontaneous ordering; ensemble phase locked to the drive indicates a driven response, and "spontaneous" vocabulary is forbidden for that result (H-13).

10.3 The coherence metric

From the phase-referenced analytic signal $z(t) = z_I + iz_Q$ RGCS-M.55 **[EST]** [3, 19], the windowed autocorrelation coherence is RGCS-M.56 **[EST]** (exact adaptation of the 3 Methods definition):

$$\mathcal{C}_w(t) = \mathcal{N}_w \int d\tau \left|\big[(\bar{z}\,\Pi_{t,w}) \star (z\,\Pi_{t,w})\big](\tau)\right|, \tag{25}$$

with boxcar window $\Pi_{t,w}$ and $\mathcal{N}_w$ normalizing a perfect tone to 1. Properties: $\mathcal{C}_w \in [0,1]$; pure tone $\to 1$; white noise $\to$ a small positive window-dependent baseline $b_w \approx (2\sqrt{\pi}/3)/\sqrt{N_w}$ ($= 0.0836$ for the project-default $N_w = 200$ samples), which is *measured per pipeline* and reported with every value — the ≈ 0.2 baseline of Koster et al. [3] is their pipeline's, never reused. Their 100 ns window at GHz carriers maps to the same tens-of-carrier-cycles compromise chosen per apparatus; a window-length sensitivity scan over at least three window lengths is mandatory. Figure 7 demonstrates the central metric property on the golden decaying-tone dataset: the coherence trace holds $\mathcal{C}_w \approx 0.208$ — well above b_w — while the amplitude has fallen to 1.6% of its peak. This is why every archival "no effect" branch is re-adjudicated with coherence analysis before its null is accepted (H-14), and why any null adjudicated on amplitude alone is labeled *amplitude-null, coherence-untested*.

11 Spatial anisotropy and coherence damping

11.1 Per-axis phase rates and the anisotropy scalar

Three orthogonal sensor channels $j = 1, 2, 3$ aligned to declared specimen axes yield analytic signals with instantaneous phases $\phi_j(t)$ and windowed phase rates $\Omega_j(t) = d\phi_j/dt$ RGCS-M.51 **[DER]**. The *spatial phase anisotropy* scalar is RGCS-M.52 **[EST]** (definition):

$$\sigma_\phi^2(t) = \frac{1}{3}\big[\Omega_{12}^2 + \Omega_{23}^2 + \Omega_{31}^2\big], \qquad \Omega_{jk} = \Omega_j - \Omega_k, \tag{26}$$

an exact adaptation of the cosmological shear scalar of Gan et al. [2, Eq. (2)], $\sigma^2 = \frac{1}{3}(H_{12}^2 + H_{23}^2 + H_{31}^2)$, under the substitution map $H_i \to \Omega_j$ (directional Hubble rates $\to$ per-axis instantaneous phase rates). In RGCS, Eq. (26) is a measured statistic of bench data and nothing more; it is zero iff all three channels evolve identically, permutation-invariant, and insensitive to a common rate. Following the kinematic decomposition lesson of Gan et al. [2], the isotropic mean rate $\Theta_\phi = (\Omega_1 + \Omega_2 + \Omega_3)/3$ is always reported alongside, with $\sigma_\phi^2/\Theta_\phi^2$ the dimensionless anisotropy. The companion circular-variance tensor $\Sigma_{\phi,jk}(t) = 1 - |\langle e^{i(\phi_j - \phi_k)}\rangle_w|$ RGCS-M.53 **[DER]** [18] captures relative-phase *locking* (0 = locked, 1 = uniformly random) with scalar reduction $\bar{\Sigma}_\phi$.

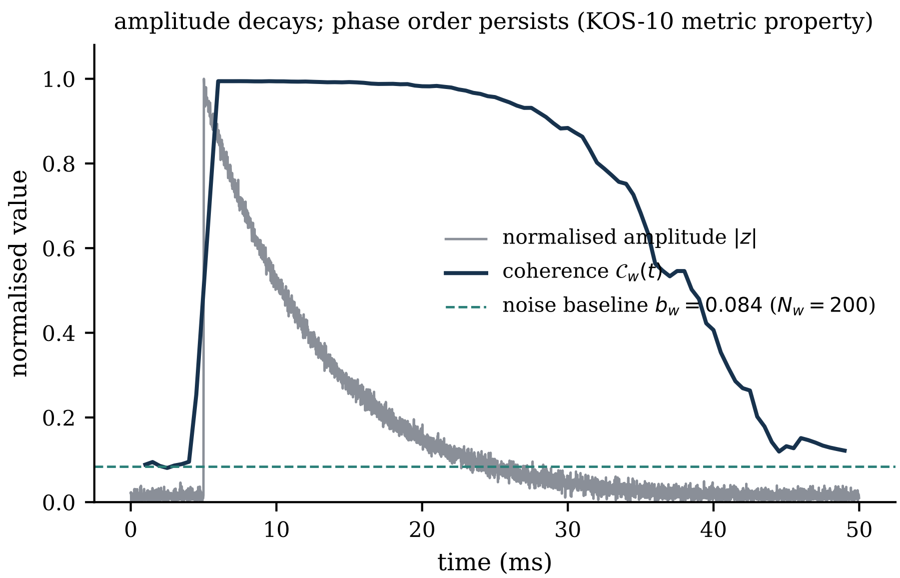

Figure 7: Coherence versus amplitude on golden dataset (c) (decaying tone in noise; Derived synthetic record). The autocorrelation coherence RGCS-M.56 persists far above the measured baseline b_w after the amplitude has decayed into the noise floor — the Koster et al. [3] metric property that motivates coherence re-adjudication of all amplitude-null archival results (H-14).

11.2 Relaxation is a model to test, not to presume

Gan et al. [2, Eq. (6)] find that after the quantum bounce their shear decays exponentially, $\sigma^2(t) \simeq \sigma_0^2\, e^{-\sigma_1 t}$ with $\sigma_1 \simeq 2.498\, m_{\mathrm{Pl}}$ — a Planck-scale, model-specific coefficient that is dimensionally meaningless for crystals and is never imported. RGCS adapts the *form* as one member of a pre-registered model set RGCS-M.54 **[HYP]** (H-10):

$$M_{\mathrm{exp}}:\ \bar{\Sigma}_\phi(t) = \bar{\Sigma}_{\phi,0}\, e^{-t/\tau_c}, \qquad M_0:\ \mathrm{const}, \qquad M_{\mathrm{pl}}:\ \bar{\Sigma}_{\phi,0}(1+t/t_0)^{-\alpha}, \qquad M_{\mathrm{se}}:\ \bar{\Sigma}_{\phi,0}\, e^{-(t/\tau_c)^{\beta_s}}, \tag{27}$$

adjudicated by AIC/BIC [20, 21] on post-drive ringdown windows, mirroring the three-model comparison layout of Gan et al. [2] (their GR / LQC / mLQC-I contrast becomes our no-decay-null / power-law / exponential contrast). The no-decay null mirrors their classical conserved-shear baseline. τ_c is a fitted bench parameter, specimen- and branch-specific; whether τ_c is *independent of drive branch* at matched energy — the bench analogue of their matter-independence result, adopted purely as a test-design pattern — is the separate Hypothesis H-11.

Figure 9 demonstrates the full pipeline on a seeded synthetic three-channel record with known exponential ground truth ($\tau_c^{\mathrm{truth}} = 60$ ms for the squared scalar): the exponential model wins with $\Delta\mathrm{AIC} = 1394$ over the no-change null and $\Delta\mathrm{AIC} = 31$ over the damped-oscillatory alternative, and the fitted $\tau_c = 61.6$ ms recovers the truth within 3%. On real data, support for H-10 additionally requires $\Delta\mathrm{AIC} \geq 4$ against the null *and* τ_c stability across same-specimen repeats, with a mandatory window-sensitivity scan.

12 Drive systems and pulse timing

RGCS keeps the three archival drive branches separate; all operating points are Source claims **[SRC]** with page provenance [6], parameterizing protocols without carrying truth value.

Electrode branch. Sharp pulses near 20 Hz (19.8 and 21 Hz as comparisons), ~15 V, silver electrodes at the selected node coordinate x_*; observables are current, voltage, charge, vibration, temperature. A nine-step progressive sequence (10–550 Hz, Fibonacci-like multipliers) is preserved together with its archival negative result (Appendix D).

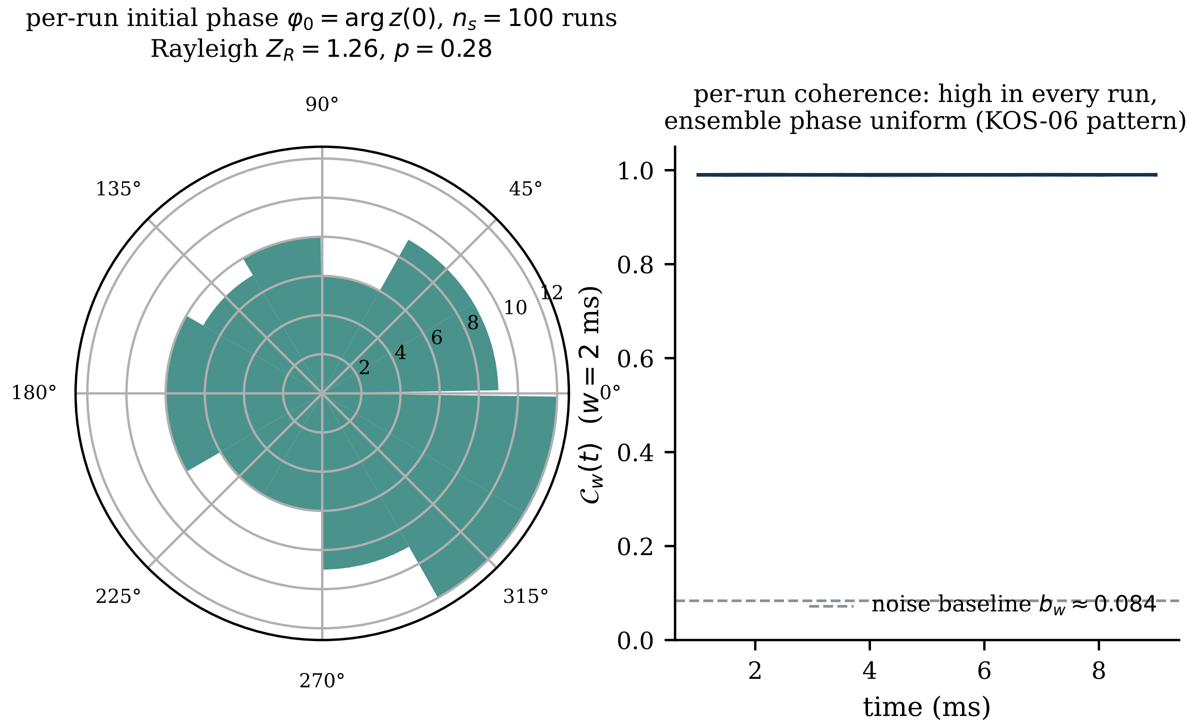

Figure 8: Phase randomization to coherence, golden dataset (d): 100 synthetic runs with seeded random initial phases. Left: ensemble initial-phase histogram with Rayleigh test ($Z_R = 1.26$, $p = 0.28$ — consistent with uniformity). The per-run initial phase is the unified estimator $\varphi_0 = \arg z(0) \bmod 2\pi$ (`rgcs_core.coherence.initial_phase_estimate`), used identically here, in Table 8, and in the body text. Right: per-run coherence traces (mean 0.990). High per-run coherence with uniform ensemble phase is the pre-registered signature pattern of spontaneous (non-imprinted) order; the companion pump-leakage dataset (f) shows the trap: ensemble uniformity rejected at $p = 6.44 \times 10^{-35}$, blocking any spontaneity claim.

Sound-key branch. Primary key 1496 Hz (644/587 Hz alternates); documented macrocycle 3 s on / 3 s off $\times 4$ $+$ 12 s pause $=$ 36 s at one-third duty.

Opposed-coil branch. Carrier 4096 Hz, complementary $180°$ non-overlap drive, pulse widths 1–4 µs, 20–60 V, copper/silver coils of 40+40 turns. The circuit inference at the archival 101 mm point (60 V, 1.3 µs, 3 A) gives $L \approx 26\,\mu\text{H}$ and $E \approx 117\,\mu\text{J}$ **[DER]** with resistance neglected — flagged "measure, don't infer": inductance is measured per coil before every campaign.

Exact-cycle envelope families. The archival millisecond prescriptions are made cycle-exact at the carrier (Table 5): the engineering refinements (e.g., the half-spacing family's $1508 = 754 + 377 + 377$ cycles over 368 ms) supplement, and never replace, the source-stated values. The phase residue $r_\phi = c - \text{round}(c)$ is defined on cycle counts only; the half-spacing nominal count 1507.328 gives $r_\phi = +0.328$ cycles. Parametric bookkeeping follows Koster et al. [3]: drive frequency, half-frequency response channel, and observed band are three distinct recorded fields, and any response at exactly half the drive frequency is logged as a parametric signature with its threshold.

13 Experimental protocols and controls

13.1 The common measurement contract

Every branch inherits a binding contract (protocol §0), modeled on the measurement chain of Koster et al. [3]:

1. **Shared reference clock.** All generation and detection instruments lock to one reference; the derivation is recorded per run. Phase or coherence claims from unshared-clock runs are non-compliant.

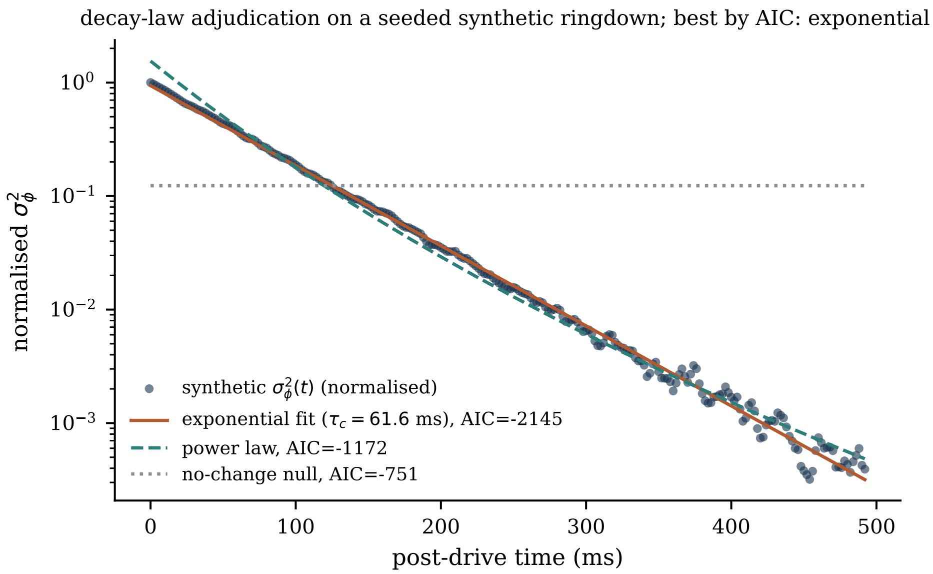

Figure 9: Anisotropy-decay model comparison RGCS-M.54 on a seeded synthetic three-channel ringdown with known exponential ground truth (Derived synthetic record; analysis entirely by `rgcs_core.coherence`). The layout follows the three-model comparison of Gan et al. [2]; the exponential form is adapted from their Eq. (6) as a fit form only, and τ_c is a bench parameter.

Table 5: Opposed-coil envelope families at the 4096 Hz carrier (RG-12). Millisecond values are Source claims from the JH log; the exact-cycle allocations are Derived engineering refinements that do not replace the source-stated values. The phase residue r_ϕ is defined on cycle counts only (D-13).

family	ON/spacing/pause (ms)	macro (ms)	duty	nominal cycles	exact allocation	r_ϕ
standard	46 / 46 / 184	552	0.3333	2260.992	2261 = 754+754+753	-0.008
half spacing	46 / 23 / 92	368	0.5000	1507.328	1508 = 754+377+377	+0.328
double rate	23 / 23 / 92	276	0.3333	1130.496	1131 = 377+377+377	+0.496

2. **Single-shot I/Q.** No averaging across excitation cycles before coherence analysis; each shot is stored individually.
3. $N \geq 100$ **runs** before any ensemble-phase statement (Koster et al. used 1,000; N is recorded per campaign).
4. **Acquisition** $\geq 2.5\times$ **past drive-off** for any coherence claim — post-drive evolution can carry the strongest ordering signal; runs below the ratio are amplitude-only.
5. **Controls precede attribution.** Instrument-only and no-specimen controls at identical settings must be complete before any coherence or enhancement claim (Figure 10).
6. **Coherence triplet.** Every coherence value ships as $(\mathcal{C}_w, w, b_w)$, always alongside an amplitude report.
7. **Sensor geometry per run.** Position and aperture of every sensor are first-class metadata; the aperture is a spatial low-pass filter on mode shapes.
8. **Artifact register.** Known instrument artifacts (frequencies, conditions) are listed per campaign and referenced from every manifest.
9. **Thresholds are apparatus-specific.** No threshold from any other apparatus — in particular the 21 dBm value of Koster et al. [3] — is ever a bench target or comparison point.
10. **Vocabulary.** Multi-channel disagreement metrics are *spatial phase anisotropy*; no condensation or dark-sector vocabulary for RGCS quantities.

Calibration gates precede every hypothesis test: instrument phase-drift budget; coherence baseline b_w with window-sensitivity scan; known-weak-tone injection establishing amplitude and coherence detection floors (which makes null claims quantitative); dated sensor calibration; and correction-ledger stability. Statistics are estimation-first — effect sizes G_c and d_c with seeded bootstrap confidence intervals [15] — with verdict-style tests pre-registered at $\alpha = 0.05$ under per-family Benjamini–Hochberg control [22], blinded analysis with

Table 6: Pre-registered experimental branches (EXPERIMENT_PROTOCOL v1.0.0). Every branch inherits the §0 contract: shared reference clock, single-shot I/Q, $N \geq 100$ runs before ensemble-phase claims, acquisition $\geq 2.5\times$ past drive-off, controls before attribution, coherence triplet reporting, per-run sensor geometry, artifact register, apparatus-specific thresholds.

branch	hypotheses	primary observables	key controls
1 modal survey	H-01, H-01a, H-03, H-04, H-07, H-09 (partial)	f_n, Q_n, γ_n, $\hat{x}_e$, parity phases	fixture-only; glass dummy; reference resonator
2 electrode pulse	H-14 (primary), H-12, H-13, H-11	G_c, d_c, $(\mathcal{C}_w, w, b_w)$, Z_R, τ_c	shielded RC dummy; dummy electrodes; sham; injection
3 sound key	H-02 (primary), H-12, H-14, H-11	ϵ-resolved d_c; key vs matched off-key	speaker-only; matched-SPL off-key; muted sham
4 opposed coil	H-14 (primary), H-09, H-10, H-12, H-13, H-11	mode-band I/Q; ring-up/ringdown envelopes; τ_c	no-crystal coils; resistor dummy; rotated coil
5 human loading	H-08, H-08b	k_H per mode; repeatability variance; ΔM_H	fixture/surrogate loads; calibrated masses; ethics gate
6 spiral cone	H-06, H-06a, H-05	apex-band G_c, d_c; fitted κ_χ vs priors	disk, plain cone, Archimedean, flat log spiral
7 water	none (exploratory)	pre/post chemistry deltas; blinded photo scoring	sham; thermal; evaporation; crystal-absent; ultrasonic
8 spatial mapping	H-03, H-10, H-11, H-13	$\sigma^2_\phi(t)$, Σ_ϕ, $\bar{\Sigma}_\phi(t)$; PLV	rigid dummy bar; common injection; channel scrambling

hash-committed freezes, and scrambled-label nulls against look-elsewhere effects.

13.2 The eight branches

Table 6 summarizes the eight pre-registered branches; Figure 10 shows the control matrix. Three boundaries are absolute. **Human loading (Branch 5) is a non-clinical bench procedure**: a volunteer holds an undriven crystal; ethics review (or documented exemption) is required before any participant contact; no physiological measurement, health-related claim, or therapeutic framing is permitted in any artifact; no drive is ever energized during human contact (hardware-interlocked). **The water branch (Branch 7) is exploratory**: it is pre-registered as estimation-only adversarial replication, tied to no H-xx hypothesis; any positive finding generates a new pre-registration and can never retroactively support existing hypotheses. **Control-subtracted metrics carry the evidence**: the fractional gain $G_c = \max(0, (\bar{Y}_{\text{cfg}} - \bar{Y}_{\text{ctl}})/\bar{Y}_{\text{ctl}})$ and standardized effect size d_c RGCS-M.57 **[EST]** are computed against dummy-load, no-crystal, and detuned ($\epsilon = 1.25$ convention) controls; the engineering merit $S = |\Lambda|^2 D_f P_\phi N_x G_c$ RGCS-M.58 **[DER]** ranks configurations for replication priority and is never evidence.

14 Worked examples generated from code

Every number in this section (and throughout the manuscript) is computed at build time by `rgcs_core` 2.0.0 [4] via `tools/make_figures.py` and `tools/make_tables.py`; nothing is hand-copied. Table 7 reproduces the project's golden reference values; Table 8 re-analyses the golden coherence datasets from their CSVs.

Example 1 — exact pair resonance. $f_x = 20480\,\text{Hz}$, $p = 2$, $f_m = 40960\,\text{Hz} \Rightarrow \epsilon_R^{(f)} = 0$ exactly. The experimental question is then whether two measured families actually occupy those frequencies, exhibit nonzero overlap Λ, and show splitting or control-surviving transfer gain when swept through the condition (H-02); Table 3 shows how quickly the class degrades with detuning at realistic Q.

Example 2 — the ambiguous 116 mm specimen. Section 4 and Table 1: $N_{\text{eff}} = 6.6402$, class set $\{6, 7\}$. RGCS treats this as a calibration problem — wave-speed anisotropy, mixed modes, fixture stiffness, and measured peaks decide the assignment, not proximity arithmetic.

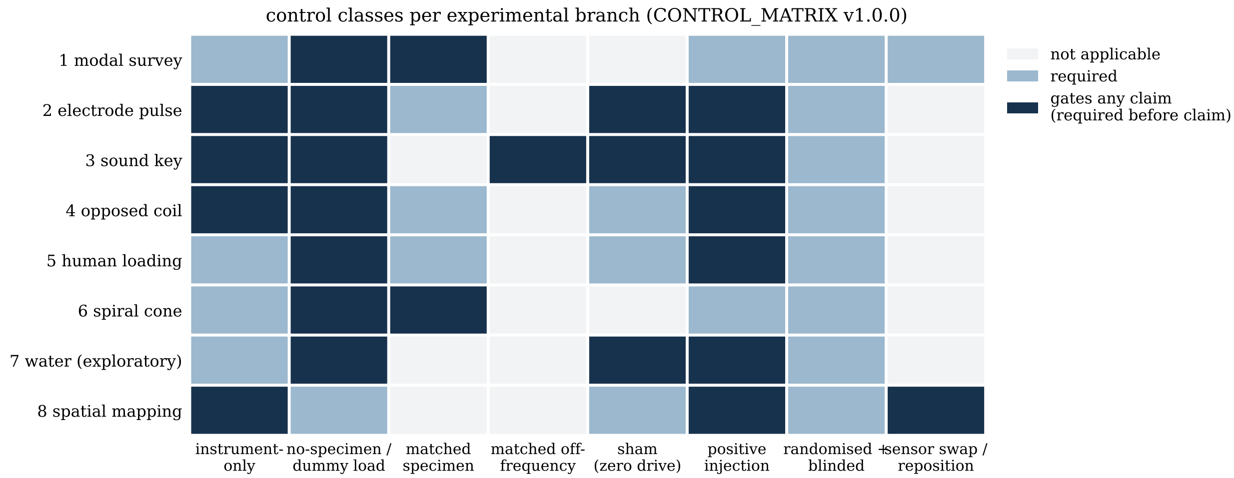

Figure 10: Experimental control matrix. Dark cells are *required-before-claim* gates: an analysis on that branch fails QA if the control is not complete at identical settings. The matched-SPL off-frequency class is the decisive test of any "key" claim; positive injection establishes the detection limits that make null results quantitative.

Example 3 — spiral prior versus placeholder. The spiral default geometry gives $\ell_{3D} = 384.022$ mm and $R_\chi^{(s)} = 15.280$ mm, hence a prior slope $\kappa_\chi \approx 65725$ Hz — versus 10042.68 Hz for the 100 mm placeholder. The pre-registered fit (H-06a) asks whether either spiral prior (mean or per-turn) falls within the κ_χ interval fitted from measured spacing, against the scrambled-index null.

Example 4 — loading. $k_H = 0.9866751$, $\Delta M_H = 2.0938$ g at $\eta = 0.5$ (Section 9) — quotable only with the η caveat or after the two-loading protocol.

Example 5 — dynamic pipeline on synthetic ground truth. Figures 7–9 and Table 8: the coherence metric recovers unity on a pure tone ($\min \mathcal{C}_w = 1.000000$), sits at the predicted baseline on white noise ($\langle \mathcal{C}_w \rangle = 0.0886$ vs $b_w = 0.0836$), detects order below the amplitude floor, distinguishes spontaneous from drive-imprinted ensembles, and the decay-law comparison recovers a known τ_c within 3%. These are software verifications [DER] — necessary before first use on data, evidential about nothing physical.

Table 7: Golden reference values (evidence-ledger Part E), recomputed at build time by `rgcs_core` 2.0.0. Ladder and compact values carry the $\pm u_v$ systematic band of RGCS-M.10.

ID	quantity	value (live)	label
G-01	ladder constant $v_L/(2f_0)$	770.263671875 mm $\pm$5%	Derived
G-02	L_5	154.052734375 mm $\pm$5%	Derived
G-03	L_7	110.037667411 mm $\pm$5%	Derived
G-04	$f_{ax}(110\,\text{mm})$	28681.8182 Hz $\pm$5%; $+9.81818$ Hz $= +0.03424\%$ vs $7f_0$	Derived-from-Hypothesis
G-05	$f_{ax}(116\,\text{mm})$; class set	27198.2759 Hz; $N_{\text{eff}} = 6.640204$; $\mathcal{N} = \{6, 7\}$	Derived
G-06	compact term n=1 (v_χ=$\bar{v}_L$, R_χ=100 mm)	10042.6769091 Hz $\pm$5%	Hypothesis-conditioned
G-07	$\epsilon_R^{(f)}(40960, 20480, p{=}2)$	0.0 (exact)	Derived
G-08	hybrids for f_A=f_B=1000 Hz, g=10 Hz	990 / 1010 Hz	Established
G-09	$k_H(152/154.052734375)$; $\Delta M_H(154\,\text{g}, \eta{=}0.5)$	0.9866751189; 2.0937873 g	Hypothesis-conditioned
G-10	$a(q{=}e)$; $r\kappa$	0.15915494309; 0.98757049215	Established
G-11	arctan $\sqrt{\varphi}$ vs 51.843°	51.8272924°; $\Delta = -0.015708°$	Established arithmetic
G-12	exact-cycle drive families (cycles)	2261 / 1508 / 1131	Derived

Table 8: Golden coherence datasets re-analysed at build time with `rgcs_core.coherence` ($w = 2$ ms, $N_w = 200$, hop 0.5 ms; $b_w = (2\sqrt{\pi}/3)/\sqrt{N_w} = 0.0836$). Per-run initial phases are the unified estimator $\varphi_0 = \arg z(0) \bmod 2\pi$ (`initial_phase_estimate`). All datasets are Derived synthetic records with known ground truth; none are measurements.

case	recomputed metrics	lesson
(a) circular white noise	$\langle \mathcal{C}_w \rangle = 0.089$ vs theory $b_w = 0.084$	baseline, not zero
(b) pure tone 5 kHz	$\min \mathcal{C}_w = 1.0000$; PL $= 1.0000$; $\langle f_{\text{inst}} \rangle = 5000.0$ Hz	unity coherence
(c) decaying tone in noise	late-window $\langle \mathcal{C}_w \rangle = 0.208 > b_w$ while amplitude $\to$ 1.6% of peak	order outlives amplitude
(d) 100 random-phase runs	per-run $\langle \mathcal{C}_w \rangle = 0.990$; ensemble Rayleigh $Z_R = 1.26$, $p = 0.28$	spontaneous-order signature
(f) pump-leakage trap	ensemble Rayleigh $p = 6.4 \times 10^{-35}$ (uniformity rejected): drive-imprinted, Stage-III claim blocked	driven, not spontaneous

ID	quantity	value (live)	label
G-13	coil inference (60 V, 1.3 µs, 3 A)	$L \approx 26$ µH; $E \approx 117$ µJ	Derived (resistance neglected)
G-14	electron-count correction (10^{-14} A)	62,415.09 e/s (not 2,000)	Established arithmetic
G-15	shear-scalar identity $\sigma_\phi^2(\Omega, \Omega, \Omega)$	0.0 (exact)	Established
D-13	phase residue $r_\phi(1507.328)$	+0.328 cycles	Derived

Figure 11 assembles the complete forward hierarchy on one page.

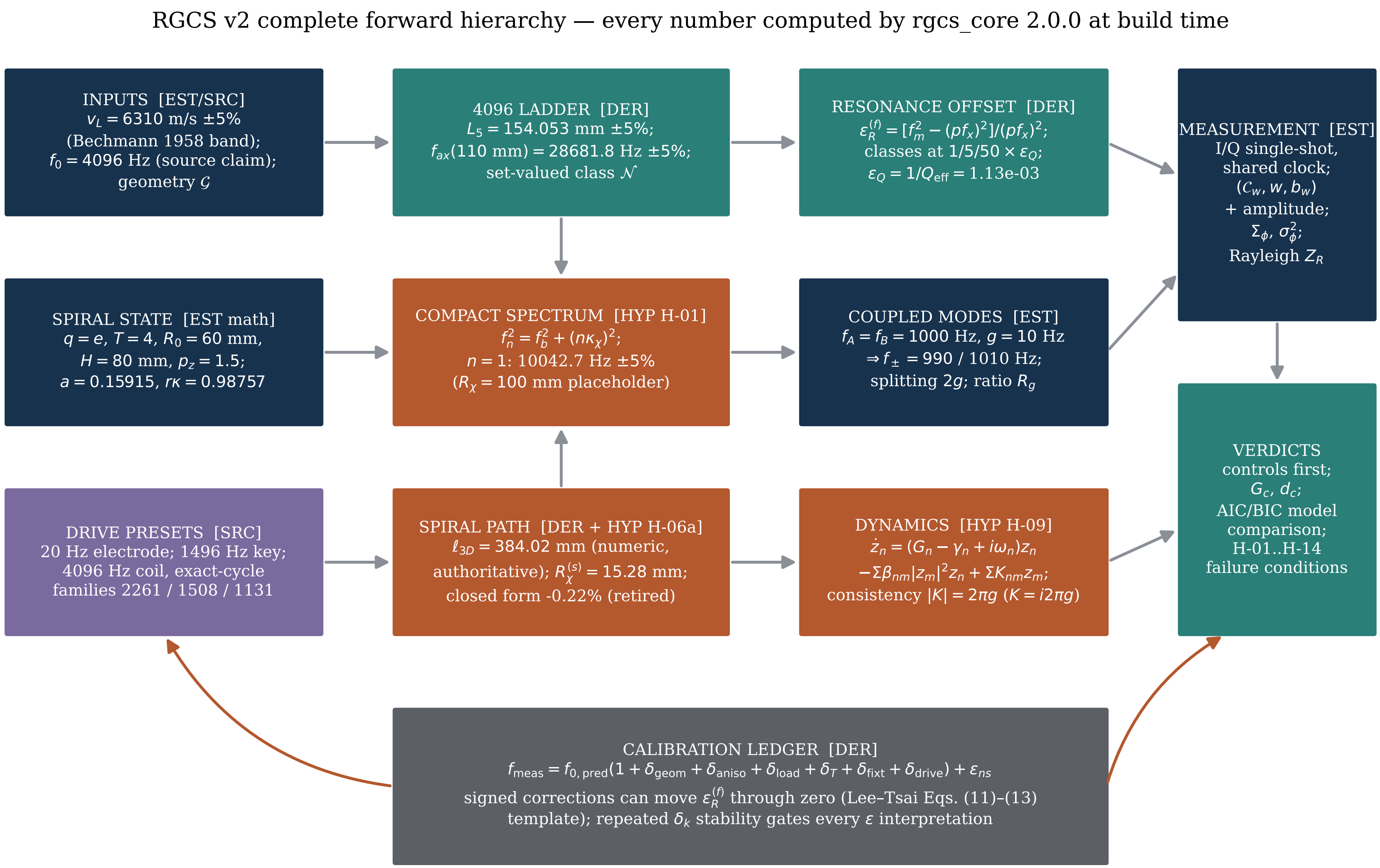

Figure 11: The complete RGCS v2 forward hierarchy, from declared inputs through spectra, interaction models, and dynamics to controlled verdicts, with the calibration ledger closing the loop. Every numerical value shown was computed by `rgcs_core` when this figure was generated; colors follow the classification tags of Section 1.

15 Falsification roadmap

Every Hypothesis-classified element ships with an observable, controls, a pre-registered failure condition, and its dominant uncertainty. All fourteen start *untested*; H-14's inputs are archival negative results. The roadmap is binding: no post-hoc bin, window, or criterion changes.

Table 9: Pre-registered hypotheses and failure conditions (abridged from the falsification roadmap; protocol branches in Table 6).

ID	hypothesis	pre-registered failure condition
H-01	compact spectrum $f_n^2 = f_b^2 + (n\kappa_\chi)^2$ describes a measured family	fails to beat linear ladder, plate law, and per-mode fit on held-out RMS/AIC, or survives index scrambling no better than chance
H-01a	1-D half-wave applies to faceted crystals	majority of specimens outside $f_{ax}(1 \pm u_v)$
H-02	enhancement localizes near $\epsilon_R^{(f)} = 0$	no localized d_c increase in the within-linewidth band vs the far band at pre-registered α
H-03	parity families are measurable and boundary-derived	assignments not reproducible across sensor placements, or indistinguishable from a random partition
H-04	zero-mode structure matches the declared flag	measured symmetric family contradicts the flag (records the Lee–Tsai analogy as partial)
H-05	coupling scales as $g \propto (2\pi R_\chi)^{-1/2}$	measured ratio differs from $\sqrt{R_2/R_1}$ beyond 2σ
H-06	spiral cone outperforms matched controls	fails to beat *every* matched control (disk, plain cone, Archimedean, flat log spiral) on the single pre-registered observable
H-06a	the spiral path is the compact path	neither spiral prior falls in the fitted κ_χ interval, or the per-mode fit wins without penalty
H-07	functional node sits at the shaft-midpoint prior x_g	$\lvert\hat{x}_e - x_g\rvert > 2\sigma_x$ reproducibly ($\hat{x}_e$ then supersedes regardless)
H-08	loading is first-order added modal mass	shifts not repeatable within tolerance; or $k_H > 1$ observed; or ΔM_H inconsistent across modes
H-08b	$\tilde{k}_H \approx k_H$	$\lvert\tilde{k}_H - k_H\rvert > u(k_H)$ — the proxy is retired to a design heuristic
H-09	amplitude–phase dynamics (22) describe ring-up, saturation, interaction	linear null fits saturation data as well ($\Delta\text{AIC} < 2$); or $\lvert K_{nm}\rvert \neq 2\pi g_{nm}$ beyond 2σ
H-10	anisotropy relaxes exponentially with time constant τ_c	exponential not preferred over the no-change null by $\Delta\text{AIC} \geq 4$, or τ_c unstable across repeats
H-11	τ_c is drive-branch independent at matched energy	branch values differ beyond combined fit uncertainty — τ_c reclassified as a calibration term
H-12	coherence shows threshold-then-saturation vs drive power	no reproducible change point across the available range (day-replication required for any threshold claim)
H-13	ordered response is spontaneous, not drive-imprinted	ensemble phase locks to drive phase (Rayleigh $p < \alpha$) — the response is driven
H-14	coherence re-adjudication of archival amplitude-nulls	if coherence is also null at the measured detection limit, the negative results upgrade to "amplitude-null AND coherence-null"; either outcome replaces the current incomplete label

What is *not* a hypothesis: the significance of $f_0 = 4096\,\text{Hz}$ (Source claim; no experiment here can confirm numerological significance); Established mathematics (verified by golden-value unit tests, not experiments); and the source presets themselves (they parameterize protocols, carrying no truth value).

16 Limitations

1. **Every hypothesis is untested.** This manuscript contains no supported physical claim beyond Established mathematics and measurement technique; it is a framework plus a test programme.

2. **The 1-D modal models are first-order.** Real quartz modes are coupled, anisotropic, and sensitive to cut, mounting, surface state, and electrodes [23–26]; the N-mode eigenproblem is a near-degenerate weak-coupling model, not elastodynamics. Finite-element and impedance-spectroscopy validation is the intended next stage.
3. **The wave-speed band dominates.** At $u_v = 0.05$ every length–frequency statement is a $\pm 5\%$ interval; harmonic classes are set-valued; no sub-percent match can mean anything until specimen orientation is measured and u_v shrinks.
4. **Identifiability limits are structural.** (v_χ, R_χ) degenerate to κ_χ; $(\eta, \Delta M_H)$ degenerate in single-loading data; (G_n, γ_n) separable only with independent ringdowns; (β_{nm}, K_{nm}) require power sweeps plus two-tone data.
5. **The dynamic model is a normal form.** Eq. (22) is the weakest structure producing threshold, locking, and ringdown; it is not a kinetic equation and makes no thermalization claim.
6. **Analogies are mathematical only.** The three papers supply equations and methods; their physics is not imported, and several of their central concepts are deliberately not adapted (Appendix D).
7. **Pilot-scale statistics.** All repeat counts are honest precision arguments, not confirmatory power claims; no prior effect sizes exist for any RGCS hypothesis, so first campaigns estimate the variances that future power calculations will consume.
8. **Source corpus quality.** The archival corpus is non-peer-reviewed, contains arithmetic errors (some corrected here as worked examples), and its claims enter only as provenance-tracked Source claims.

17 Reproducibility and software

The computational core `rgcs_core` 2.0.0 [4] is a deterministic, typed Python library (numpy/scipy/pydantic) implementing equations RGCS-M.0–M.61 with: classification metadata on every claim-bearing public function (machine-checked); a forbidden-vocabulary lint; JSON serialization in which NaN/∞ always become `null`; seeded randomness only; and unit conversions appearing exactly once per function. The test suite comprises 227 automated tests: 118 unit tests, 17 property-based suites (spectrum monotonicity, eigenvalue interlacing, circular-statistic bounds, shear-scalar invariances), 19 golden-value tests reproducing Table 7, 50 regression tests (every expected value of the golden coherence manifest recomputed from the CSVs, generator determinism, the corrected time–frequency coupling map), plus 10 desktop integration tests and 13 UI smoke tests.

This manuscript is itself a reproducibility artifact: `tools/make_figures.py` and `tools/make_tables.py` import `rgcs_core` from the repository, regenerate every figure (vector PDF), every table, and every inline numerical macro used above, and the document is compiled with `latexmk -xelatex`. Reviewing agents are expected to recompute all examples. Experiment data schemas (run manifest, specimen, environment, analysis result), the control-matrix JSON, branch templates, and the pre-registered statistical analysis plan live alongside the code; analyses are re-runnable from manifests alone. CAD generators [27, 28] embed the same constants with classification-labeled headers.

18 Conclusion

RGCS v2 turns a source-audit programme into one coherent eigenmode, dynamics, and calibration system. The 4096 ladder supplies an axial design family carried honestly inside its $\pm 5\%$ wave-speed band **[DER]**. The spiral supplies a scale-rotation path whose authoritative numeric length feeds two competing compact-radius priors **[DER]**/**[HYP]**. Lee and Tsai's architecture supplies — as mathematics, with the substitution map stated at every use — a compact-mode spectrum, parity families with an explicit zero-mode flag, a signed resonance offset with corrections that can move it through zero, and a coupling-scaling test **[HYP]** [1]. Coupled-mode theory supplies hybrid eigenvalues and the avoided-crossing signature **[EST]**. Gan et al. supply the anisotropy scalar and the decay-law adjudication methodology [2]; Koster et al. supply the phase-referenced single-shot architecture, the coherence metric and its triplet reporting rule, and the spontaneity test [3] **[EST]**. The JH audit supplies drive branches, node-placement requirements, source presets, and preserved negative results **[SRC]**. Fourteen pre-registered hypotheses with failure conditions, eight control-gated experimental branches, and a build system in which every number is regenerated from the tested core give the programme its verdict mechanism. The final word belongs to control-subtracted data: *physical correspondence is determined by measurement, not assumed by terminology.*

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A Notation and units (frozen)

Single-owner base letters (never reused bare): p (pair multiple), a (spiral pitch), Q (quality factor), S (merit score), q (spiral growth ratio), ρ (density), L (crystal length), g (two-mode coupling), n (compact/parity index), N (axial harmonic index). Bare σ is forbidden (σ_x, σ_ϕ^2, σ_v are the registered forms). Frequencies in Hz, angular frequencies in rad/s, geometry in mm, wave speeds in m/s, masses in g, time in s; unit conversions appear explicitly in equations.

Table 10: Principal symbols (abridged from the frozen notation table).

symbol	definition	unit	label
L, D_w, D_n, N_f	crystal length; wide/narrow diameters; facet count	mm, —	[EST]
α_f, α_m	female/male termination angles (per declared convention)	deg	[SRC] (values)
h_f, h_m, h_s	cap heights; shaft length $L - h_f - h_m$	mm	[DER]
V, m, ρ	volume; mass; density (2.65 default)	cm^3, g, g/cm^3	[DER] [EST]
v_L, $\bar{v}_L$, u_v	wave speed as uncertain parameter; central value; relative uncertainty	m/s, —	[HYP] [SRC] [DER]
f_0, f_{ax}, L_N, N_{eff}, $\mathcal{N}$	ladder base; axial estimate; ideal length; continuous class; class set	Hz, mm	[SRC] [DER]
χ, R_χ, v_χ, κ_χ, f_b, f_n	compact coordinate; radius; speed; slope $v_\chi/2\pi R_\chi$; base; mode frequency	rad, mm, m/s, Hz	[HYP] [DER]
P, n	parity family $\{P^-, P^+, \text{all}\}$; mode index	—	[EST] (defn)
f_m, f_x, p, $\epsilon_R^{(f)}$, d_{lin}, ϵ_Q	bridge/target modes; pair multiple; offset; linear detuning; class scale $1/Q_{\text{eff}}$	Hz, —	[EST] [DER]
$\mathbf{H}$, g, $f_\pm$, ϑ_{mix}, Γ_n, R_g	coupled-mode matrix; coupling; hybrids; mixing angle; FWHM linewidth; strong-coupling ratio	Hz, rad	[EST] [DER]
q, a, T, R_0, H, p_z, Ω_s	spiral growth; pitch; turns; outer radius; height; cone exponent; winding	—, mm	[EST] [DER]
ℓ_{pl}, ℓ_{3D}, ℓ_k, $R_\chi^{(s)}$, $R_{\chi,k}$	planar length; authoritative 3-D length; per-turn lengths; radius priors	mm	[EST] [DER] [HYP]
x_m, x_g, $\hat{x}_e$, x_*, σ_x, N_x	metric center; shaft-midpoint prior; measured node; selected node; scan uncertainty; alignment heuristic	mm, —	[EST] [DER]
k_H, $\tilde{k}_H$, η, ΔM_H, δ_k	loading factor (measured); length proxy (distinct!); modal fraction; added modal mass; signed corrections	—, g	[EST] [HYP] [DER]
z_n, A_n, ϕ_n, G_n, γ_n, β_{nm}, K_{nm}	modal amplitudes/phases; gain; damping $\omega_n/2Q_n$; saturation; coupling rate	X, s^{-1}	[HYP] [EST]
z, z_I, z_Q, $\mathcal{C}_w$, w, b_w	analytic signal; quadratures; coherence; window; baseline	X, s, —	[EST]
Ω_j, σ_ϕ^2, Σ_ϕ, τ_c, V_j, $\bar{R}_j$	per-axis phase rates; anisotropy scalar; variance tensor; relaxation time (fitted); circular variance; resultant	rad/s, s^{-2}, s	[DER] [HYP] [EST]
Λ, r_ϕ, D_f, P_ϕ, G_c, d_c, S	overlap; phase residue (cycles); detuning factor; phase closure; control-subtracted gain; effect size; merit (never evidence)	—	[DER] [EST]
Z_R, n_s, Θ	Rayleigh statistic; shot count; fit parameter vector	—	[EST]

B Derivations

B.1 Added modal mass

For a mode with effective stiffness k_s and modal mass M_{eff}, $f \propto \sqrt{k_s/M}$; adding ΔM at the antinode with full modal participation gives $f_{\text{loaded}}/f_{\text{free}} = \sqrt{M_{\text{eff}}/(M_{\text{eff}}+\Delta M)} = k_H$, hence $\Delta M_H = M_{\text{eff}}(k_H^{-2} - 1)$ RGCS-M.43. The conditioning (pure added mass, participation η) is Hypothesis H-08.

B.2 Offset–detuning relation and class scale

With $f_m = (1+d_{\text{lin}})\,pf_x$, $\epsilon_R^{(f)} = (1+d_{\text{lin}})^2 - 1 = 2d_{\text{lin}} + d_{\text{lin}}^2$ RGCS-M.19. A pair with Q_{eff} has combined fractional half-linewidth $1/(2Q_{\text{eff}})$ in d_{lin}, i.e. $|\epsilon_R^{(f)}| \approx 1/Q_{\text{eff}} = \epsilon_Q$ at the half-linewidth — the class unit of RGCS-M.20.

B.3 Coherence normalization

For the discrete windowed autocorrelation $\text{ACF}(\tau) = \sum_n z_{n+\tau}\bar{z}_n$ over N_w samples, a perfect tone gives $|\text{ACF}(\tau)| = (N_w - |\tau|)A^2$ and $\sum_\tau (N_w - |\tau|) = N_w^2$, so dividing by $N_w \sum_n |z_n|^2$ normalizes the tone to exactly 1 — the discrete form of the Koster et al. [3] normalization against a perfectly coherent dummy signal. For circular white noise the same statistic concentrates near $(2\sqrt{\pi}/3)/\sqrt{N_w}$, the theoretical baseline quoted with every coherence value.

B.4 Time–frequency coupling consistency

For a degenerate pair coupled anti-Hermitianly, $K_{12} = K_{21} = i\,2\pi g$ in Eq. (22), the linear eigenvalues are $i(\omega_0 \pm 2\pi g)$: spectral peaks at $f_0 \pm g$ (splitting $2g$ Hz) and an amplitude beat $|z_1(t)| = |\cos(2\pi g t)|$ with node spacing $1/(4g)$. Fitted $|K_{nm}|$ and g_{nm} describing the same physics must therefore satisfy $|K_{nm}| = 2\pi g_{nm}$ within uncertainty (pre-registered warning flag otherwise). *Erratum (this revision): the previous requirement $K = \pi g$ with real-symmetric coupling was wrong in structure and magnitude — real-symmetric coupling splits growth rates, not frequencies, and would have flagged correct fit pairs as inconsistent.*

C Algorithms

Density inverse RGCS-M.7**.** Newton iteration on $\rho V(s_D) - m^*$ with analytic monotonicity guarantee, bisection fallback, convergence $|\Delta m|/m^* < 10^{-10}$.

Spiral path length RGCS-M.35**.** Polyline sum over θ with ≥ 1200 initial samples, doubling until $|\ell^{(2s)} - \ell^{(s)}|/\ell^{(2s)} < 10^{-6}$ (Richardson-checked); per-turn segmentation for ℓ_k.

Compact spectrum RGCS-M.13–M.17**.** Parity index-set generation with the zero-mode flag; mutually exclusive (κ_χ)-vs-(v_χ, R_χ) parameterizations; first-order uncertainty propagation to every mode.

N-mode eigenproblem RGCS-M.28**.** Symmetric validation (zero diagonal, $g_{nm} = g_{mn}$), `numpy.linalg.eigh`.

Stuart–Landau integration RGCS-M.46**.** Complex-form Euler–Maruyama with exponential integrating factor for the stiff linear part; seed required whenever noise > 0; amplitude and phase returned separately.

Coherence series RGCS-M.56**.** Sliding boxcar (w, hop declared), per-window FFT autocorrelation, tone-normalized; baseline theory $(2\sqrt{\pi}/3)/\sqrt{N_w}$.

Windowed phase rates / anisotropy RGCS-M.51–M.53**.** Unwrapped-phase least-squares slope per window (normative estimator); scalar and tensor reductions; common-injection validation control.

Model comparison RGCS-M.60**.** Four fixed models (exponential, power law, damped oscillatory, no-change), full AIC/BIC table always published; ties at $\Delta\text{AIC} < 2$; scrambled-label permutation nulls (≥ 1000 seeded permutations) for spectra and parity.

Threshold detection. Midpoint-crossing estimator with seeded bootstrap CI; day-replication required before any threshold claim.

D Source-claim ledger, traceability, and negative results

D.1 Adapted-mathematics traceability matrix

Table 11: Paper-to-system traceability with classification. Every adapted equation states its substitution map at point of use; "validation" names the measurement that would support or kill the row.

paper element	source role	RGCS adaptation	validation	label
Lee & Tsai [1]. *Maps* $m \to f$, $m_B \to f_b$, $1/R \to v_\chi/2\pi R_\chi$; *zero-mode flag stated at every use.*				
compact coordinate (orbifold y, radius R)	dark-sector compactification	closed wave path χ, fitted R_χ (via κ_χ)	fitted spacing, H-01	[HYP]
KK towers, Eqs. (9)–(10)	tree-level mass spectra	Eq. (10)	held-out prediction	[HYP]
BC parity, Table I, Eqs. (1)–(2)	orbifold mode selection	$P^\pm$ families + `include_zero_mode`	two-sensor phase, H-03/H-04	[EST] (math)
resonance level, Eq. (13)	mediator-pair offset	$\epsilon_R^{(f)}$, Eq. (13), same sign convention	sign-resolved sweep, H-02	[DER]
loop corrections, Eqs. (11)–(12)	signed mass shifts	signed correction ledger δ_k	ledger stability	[DER]
coupling reduction, Eq. (6)	$g_1 = g_1^{(5)}/\sqrt{L}$	$g \propto (2\pi R_\chi)^{-1/2}$	refit after geometry change, H-05	[HYP]
$\epsilon_R = 1.25$ contour	their non-resonant reference	far-detuned *control convention* only	control runs	[DER] (conv.)
Gan et al. [2] — *map:* $H_i \to \Omega_j$				
shear scalar, Eq. (2)	Bianchi-I anisotropy measure	σ_ϕ^2, Eq. (26) (bench statistic)	golden identities; common-injection control	[EST] (defn)
kinematic decomposition	expansion/vorticity/shear	mean rate Θ_ϕ reported beside σ_ϕ^2	three-channel data	[DER]
exponential decay, Eq. (6)	post-bounce shear damping	$M_{\exp}$ fit form; τ_c bench-fitted	AIC/BIC set, H-10	[HYP] (form cited)
matter independence	five matter models, same exponent	branch-independence of τ_c as test design	matched-energy branches, H-11	[HYP]
three-model comparison	GR/LQC/mLQC-I layout	null/power/exponential comparison plots	COH-M14	[EST] (method)
Koster et al. [3]				
I/Q single-shot chain	phase-referenced detection	measurement contract §13	instrument budget	[EST]
autocorrelation coherence	Methods metric	$\mathcal{C}_w$, Eq. (25), triplet rule	golden cases (a)–(c)	[EST]
1,000-shot phase ensembles	spontaneity evidence	Rayleigh Z_R protocol, H-13	$n_s \geq 100$ ensembles	[EST] (method)
threshold behavior	pump-power onset	change-point + day-replication, H-12	power sweeps	[HYP]
sub-amplitude coherence	remanence below noise floor	null re-adjudication rule, H-14	detection-limit injection	[DER] (rule)
parametric $f/2$ pumping	pair injection	bookkeeping fields; parametric branch terms (off by default)	$f/2$ response logging	[DER]
aperture spatial filtering	antenna as low-pass	sensor aperture metadata + correction concept	sensor-placement sweep	[EST]

D.2 Not adapted (binding exclusion lists)

From Lee & Tsai: relic abundance and freeze-out; kinetic-mixing values κ; axial-vector DM coupling; direct-detection cross-sections; SN1987A bounds; the invisible/visible dark-photon distinction; decay chains; small-scale-structure conclusions; the 10^{-8}–10^{-4} window as anything but source context. **From Gan et al.:** the Planck-unit coefficient $\sigma_1 \simeq 2.498\, m_{\rm Pl}$; bounce times; Barbero–Immirzi parameter; holonomy corrections; singularity resolution; any isotropization-of-the-universe claim. **From Koster et al.:** the BEC/condensation interpretation as applied to any RGCS system; magnon dispersion physics; the 281 mT / 7.8 GHz / 21 dBm operating values as targets; YIG material physics; spontaneous-symmetry-breaking claims for RGCS responses. None of these may appear as an RGCS physical claim, code default, or bench target.

D.3 Preserved negative results [SRC] (negative)

- **JH-015:** the nine-step progressive electrode sequence (10–550 Hz) at 20/40/60 V produced no detected acoustic response ("no singing"). Status: *amplitude-null, coherence-untested*; re-adjudication is Branch 2 / H-14.
- **JH-031:** plant-stimulation test at 100 V / 300 ns: no effect reported. Status: *amplitude-null, coherence-untested*.
- **JH-033:** minute charge-amplifier signals in the 101 mm coil configuration were attributed by the source to drive pickup. Status: pickup floor; the no-crystal energized-coil control of Branch 4 makes this subtraction mandatory before any coil-branch claim.

D.4 Numerology containment [EST] (arithmetic) / [SRC] (significance)

The powers-of-eight ladder ($64 \cdot 8^{n-1}$, 4096 at level 3), the 2.45 GHz ratio decoding, and the golden-angle candidates are preserved as audited arithmetic with corrections of source typos; the angle audit gives $\arctan\sqrt{\varphi} = 51.827292^\circ$ against the proposed 51.843° ($\Delta = -0.015708^\circ$, quantified and nonzero). No physical inference is drawn from any of it, and the scrambled-label null of the statistical plan exists precisely to kill nearest-integer coincidence hunting.